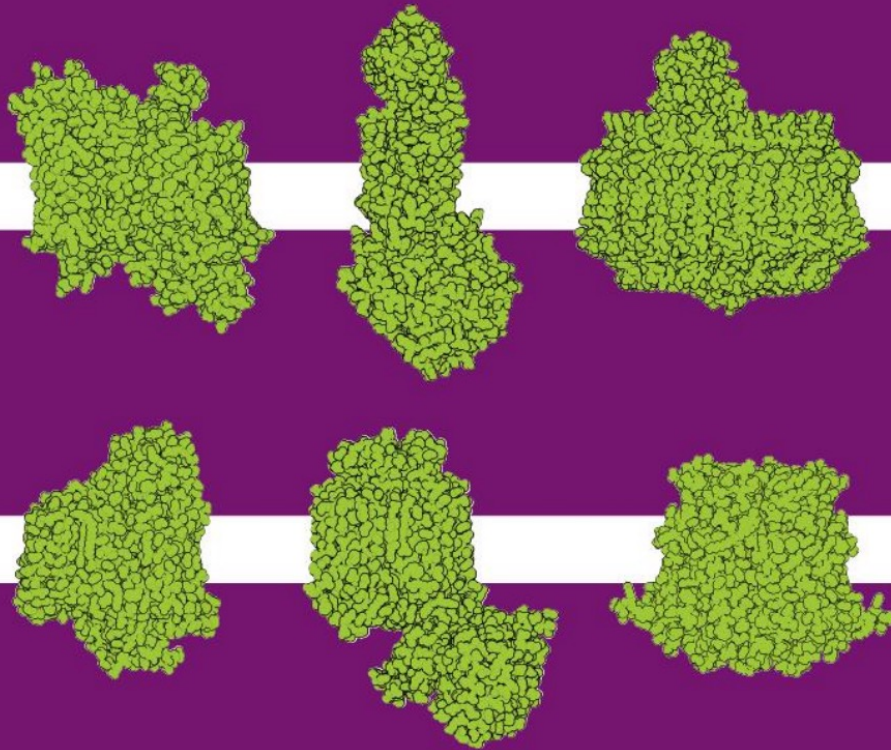


MOLECULAR BIOLOGY OF
THE CELL
SEVENTH EDITION



ALBERTS HEALD JOHNSON MORGAN RAFF ROBERTS WALTER

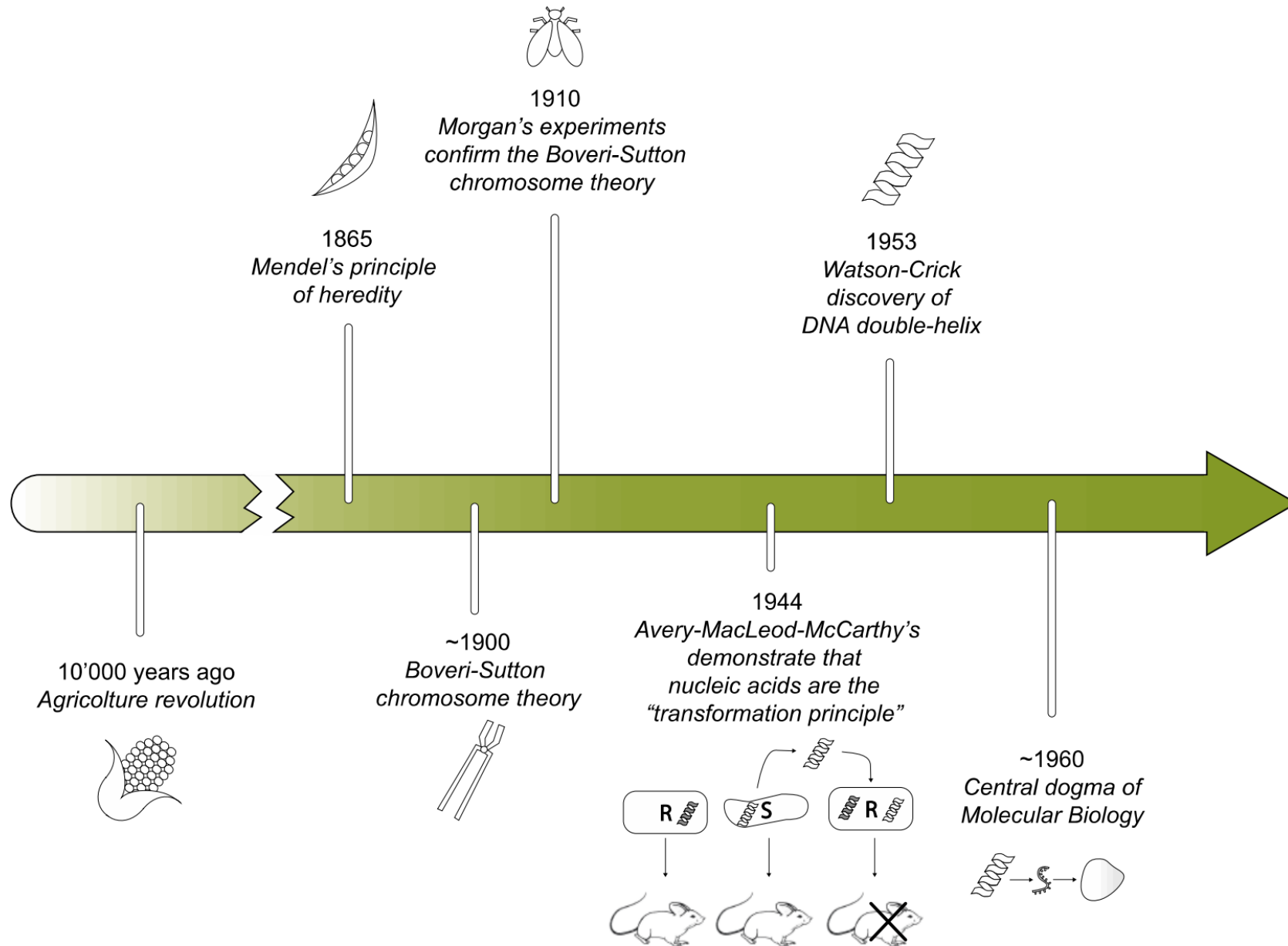
Chapter 6

How Cells Read the Genome: From DNA to Protein

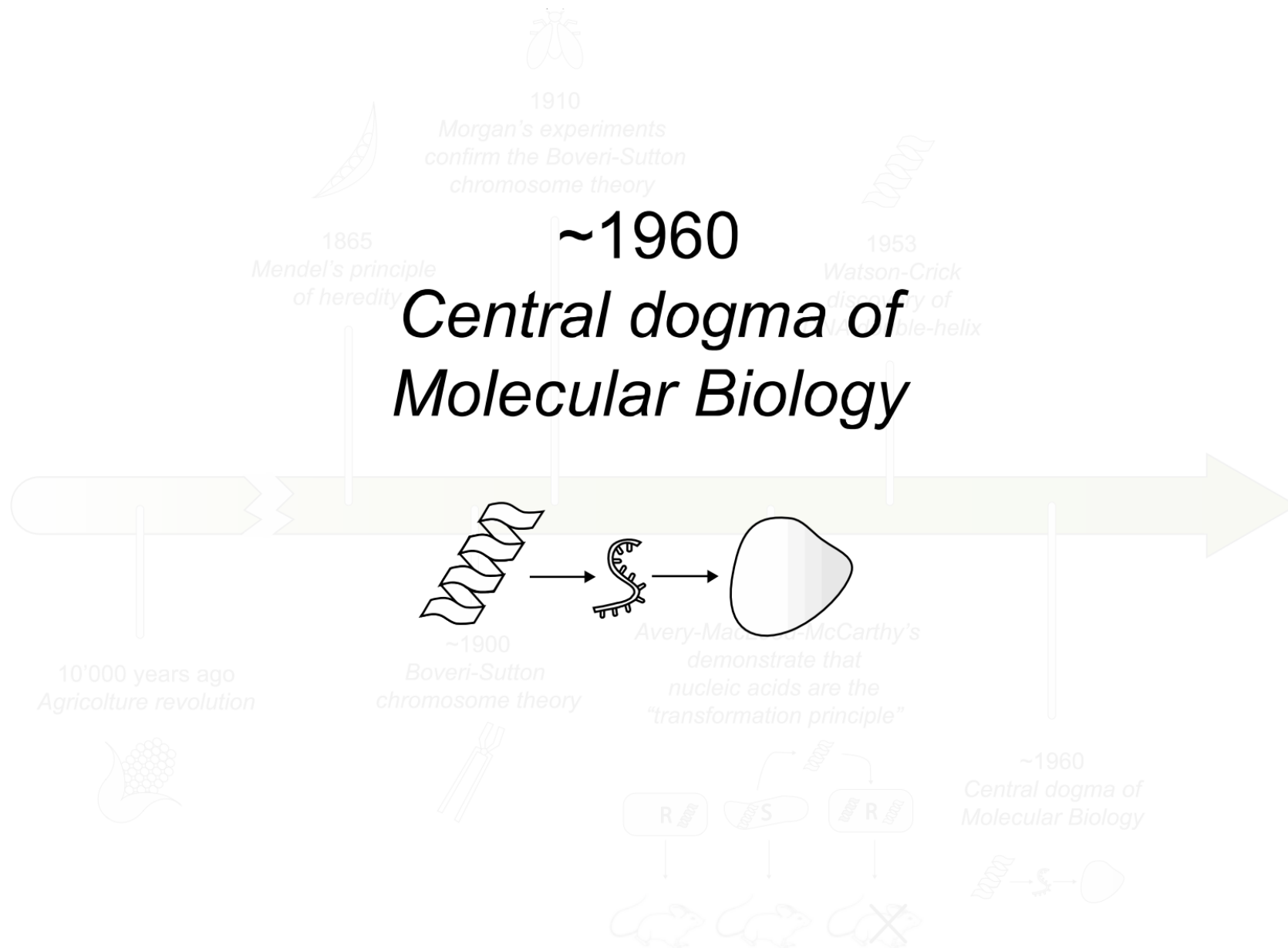
PART 1

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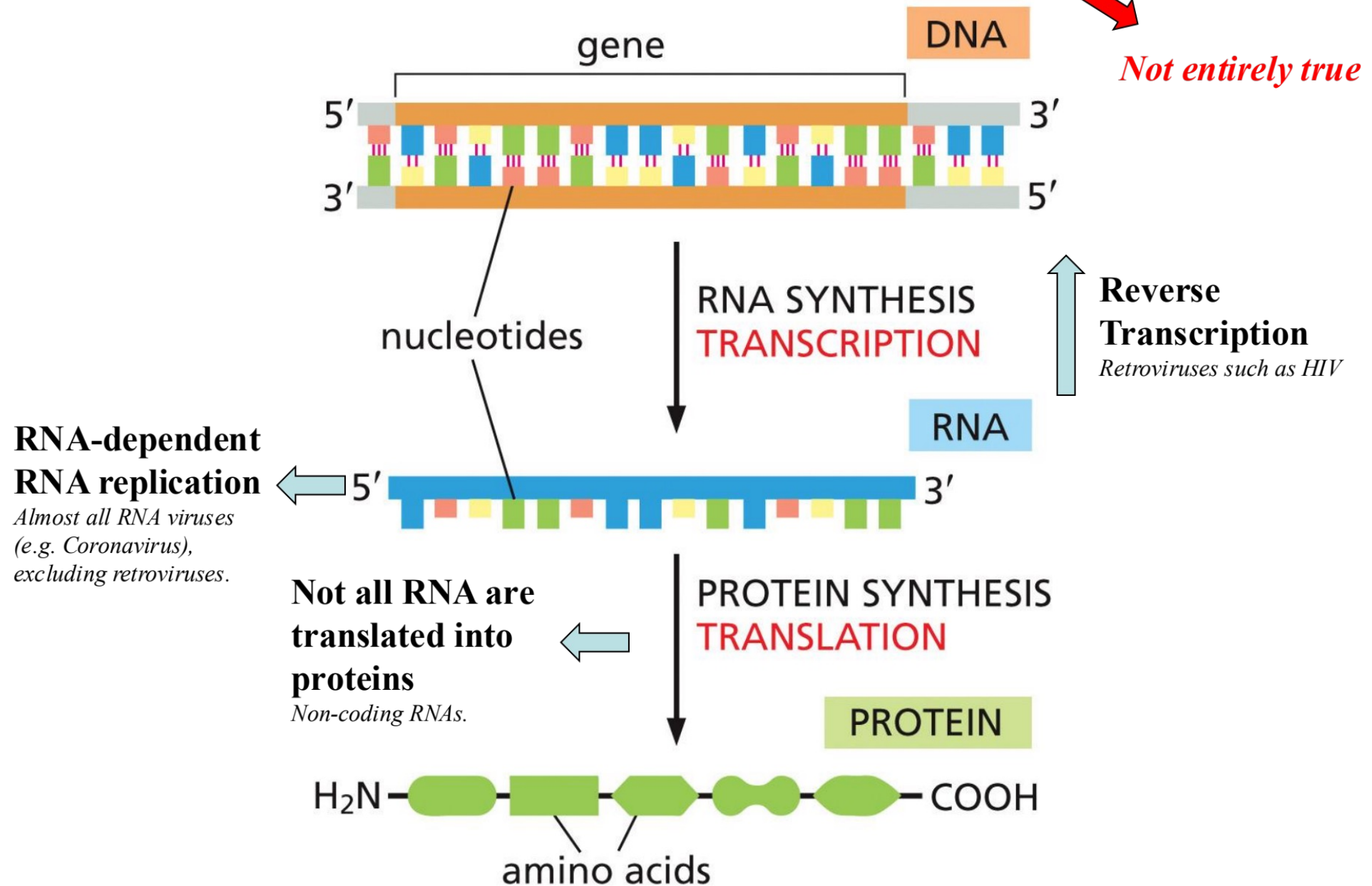
Key scientific breakthrough leading to the discovery of DNA as the molecule of life



Key scientific breakthrough leading to the discovery of DNA as the molecule of life



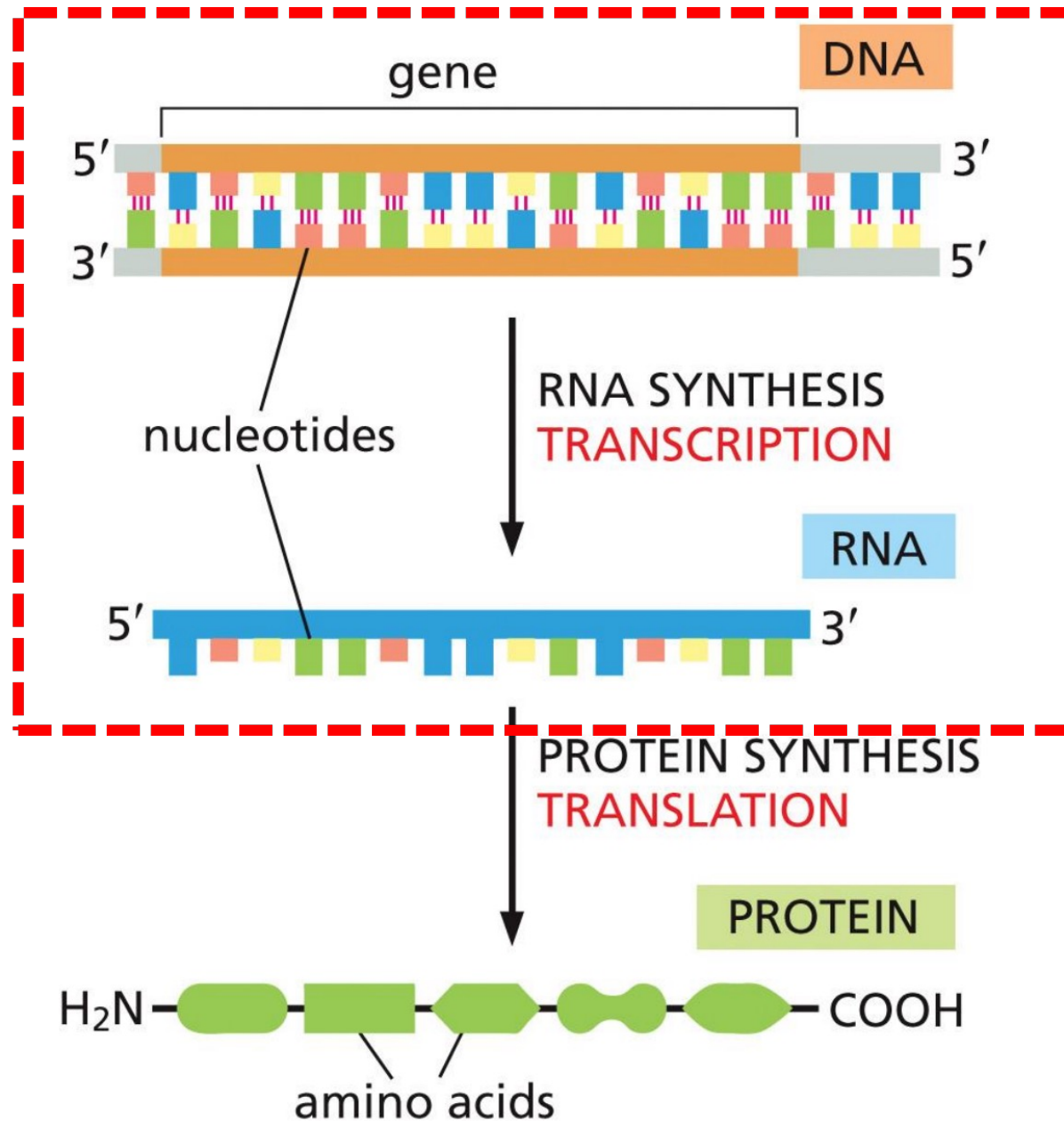
Central dogma of molecular biology: Francis Crick 1958



Central dogma of molecular biology: Francis Crick 1958

Today's lecture:

From DNA to
RNA



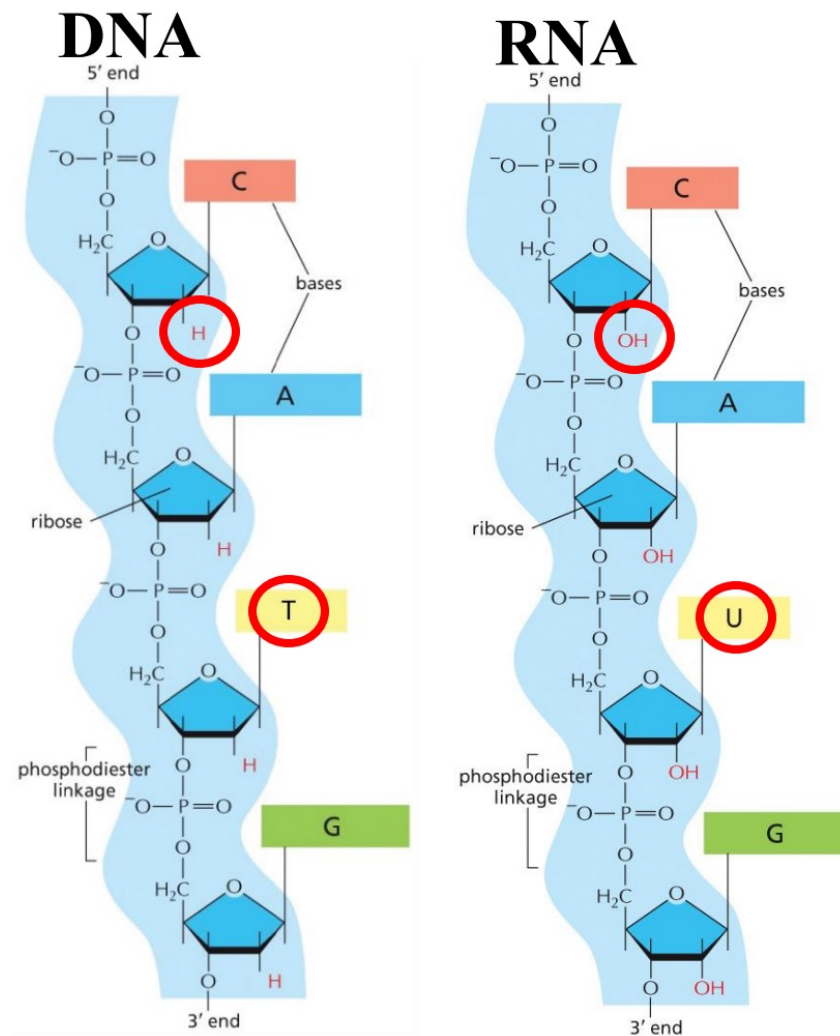
...What is RNA?

...What is RNA's function?

Until the 1950s, the role of RNA in cellular machinery was unknown:

RNA was known as a similar molecule to DNA but with slight differences, such as:

- The sugar (**ribose** in RNA vs. deoxyribose in DNA);
- One base change (**uracil** in RNA vs. thymine in DNA);



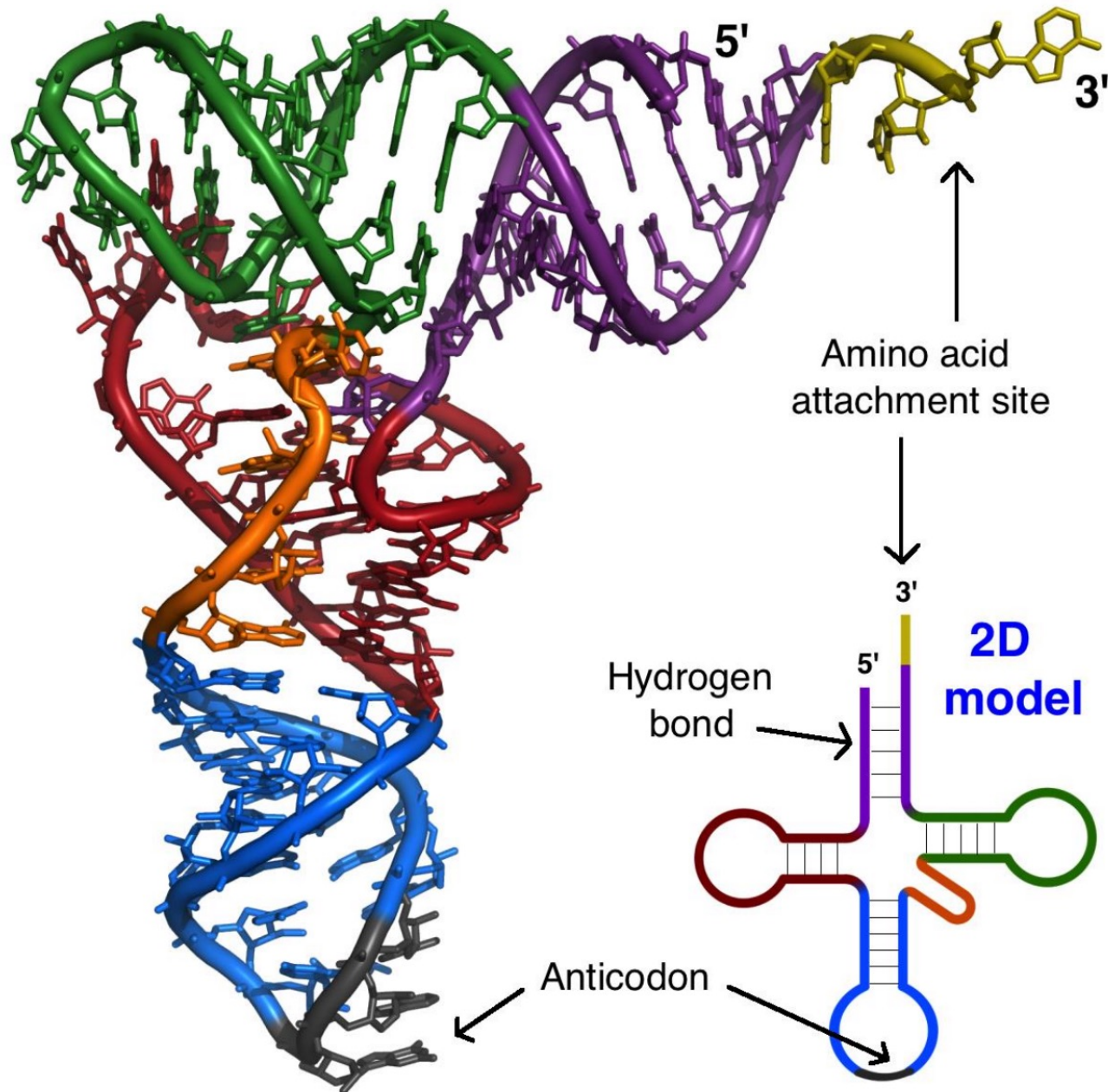
Until the 1950s, the role of RNA in cellular machinery was unknown:

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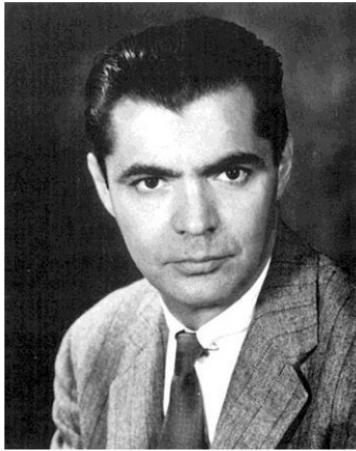
- The sugar (**ribose** in RNA vs. deoxyribose in DNA);
- One base change (**uracil** in RNA vs. thymine in DNA);



Example RNA 3D structure: tRNA

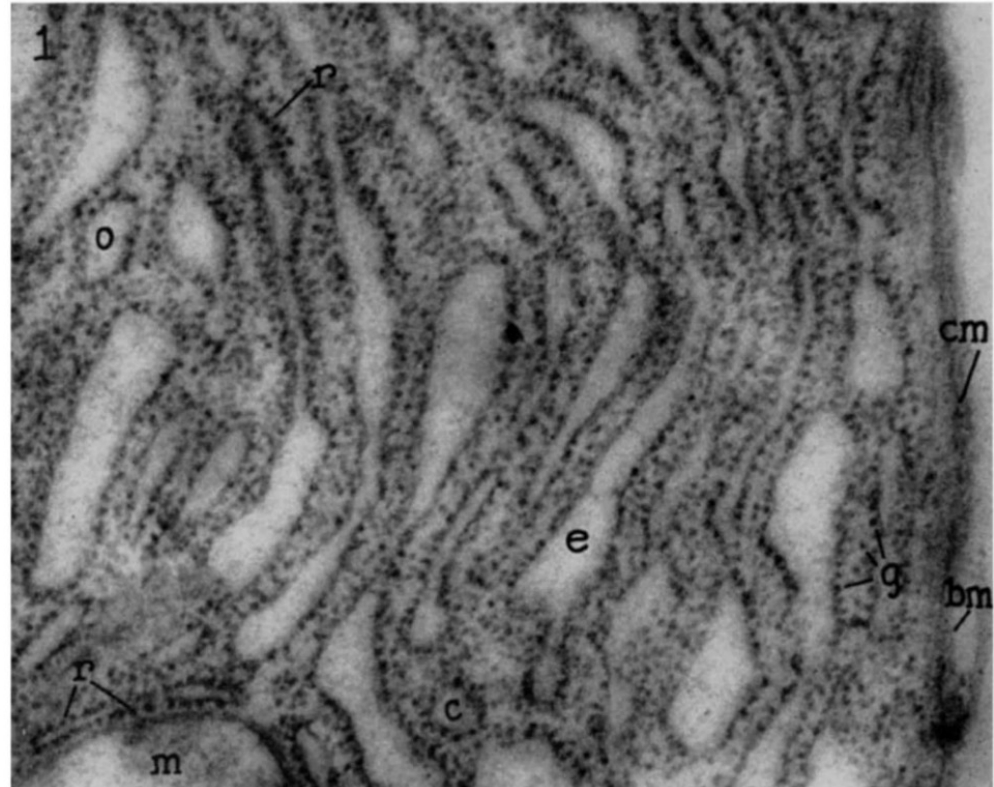


Until the 1950s, the role of RNA in cellular machinery was unknown:



George Palade

Noble Medicine and Physiology 1974



Hypothesized that ribosomes play a key role in protein synthesis. Figure: ribosomes (g) are present in high number in cells actively synthesizing proteins. Model: acinar cell of rat pancreas.

Until the 1950s, the role of RNA in cellular machinery was unknown:



**Paul Zamecnik and
Mahlon Hoagland**

Cell-free protein synthesis from rat liver:

- By introducing radiolabeled amino acids, tracked their incorporation into proteins;
- Demonstrated that ribosomes were the key site for protein assembly;
- Discovered transfer RNA (tRNA).

YOU ARE THE SCIENTIST

Is RNA the messenger between DNA and protein?

What is known: Ribosomes are the site where proteins are produced. tRNAs 'carry' aminoacids to proteins.

What is not known: where does the information for producing a given protein come from?

Which experiment/result would *best* reveal our candidate “messenger” molecule?

- A** The candidate should be very close to the DNA, how do you copy the information otherwise???
- B** If the candidate is removed, protein levels should remain unchanged
- C** If the candidate is removed, protein levels should increase
- D** An increase in protein levels should correspond to an increase in our candidate levels, and vice versa

Until the 1950s, the role of RNA in cellular machinery was unknown:



Elliott Volkin and Lazarus Astrachan

Following bacteriophage infection in bacteria, RNA levels rapidly increased. Hypothesis: RNA as a possible intermediate between DNA and protein synthesis.

Until the 1950s, the role of RNA in cellular machinery was unknown:



**Sydney Brenner, François Jacob, and
Matthew Meselson experiment**

E. Coli grown in a medium with heavy isotopes, making ribosomes "heavy". After infecting the bacteria with a bacteriophage, shifted to a medium with light isotopes. Newly synthesized mRNA incorporated the lighter isotopes. New Model: mRNA intermediate between DNA and protein

...What is RNA?

...What is RNA's function?

TABLE 6–1 Principal Types of RNAs Produced in Cells

Type of RNA	Function
mRNAs	Messenger RNAs, code for proteins
rRNAs	Ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	Transfer RNAs, central to protein synthesis as the adaptors between mRNA and amino acids

...What is RNA?

...What is RNA's function?

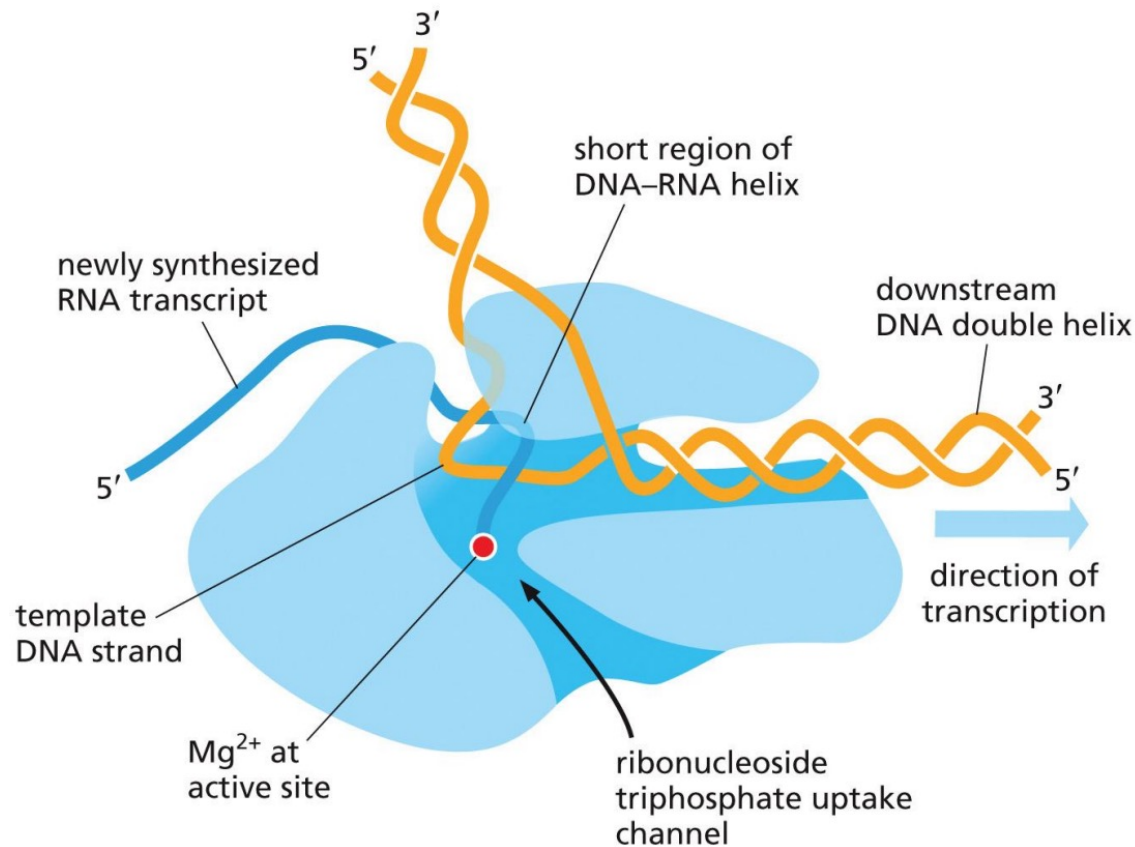
TABLE 6–1 Principal Types of RNAs Produced in Cells

Type of RNA	Function
mRNAs	Messenger RNAs, code for proteins
rRNAs	Ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	Transfer RNAs, central to protein synthesis as the adaptors between mRNA and amino acids
Telomerase RNA	Serves as the template for the telomerase enzyme that extends the ends of chromosomes
snRNAs	Small nuclear RNAs, function in a variety of nuclear processes, including the splicing of pre-mRNA
snoRNAs	Small nucleolar RNAs, help to process and chemically modify rRNAs
lncRNAs	Long noncoding RNAs, not all of which appear to have a function; some serve as scaffolds and regulate diverse cell processes, including X-chromosome inactivation
miRNAs	MicroRNAs, regulate gene expression by blocking translation of specific mRNAs and causing their degradation
siRNAs	Small interfering RNAs, turn off gene expression by directing the degradation of selective mRNAs and helping to establish repressive chromatin structures
piRNAs	Piwi-interacting RNAs, bind to piwi proteins and protect the germ line from transposable elements

...From DNA to RNA

**...In which direction is
DNA/RNA synthesis?**

Transcription



10 nm

- Transcription shares similarities with DNA replication (opening, unwinding, base pairing);
- However, key differences: RNA strand detaches, length;
- RNA polymerases are highly processive enzymes (but less than DNA pol);
- RNA polymerases have a higher error rate compared to DNA polymerases;
- RNA and DNA polymerases structurally different.

**...transcription initiation, termination
and mRNA processing differ between
prokaryotes and eukaryotes.**

Abortive initiation in bacterial transcription



Embedded video, 5:01. [Click to play.](#)

Transcription in prokaryotes

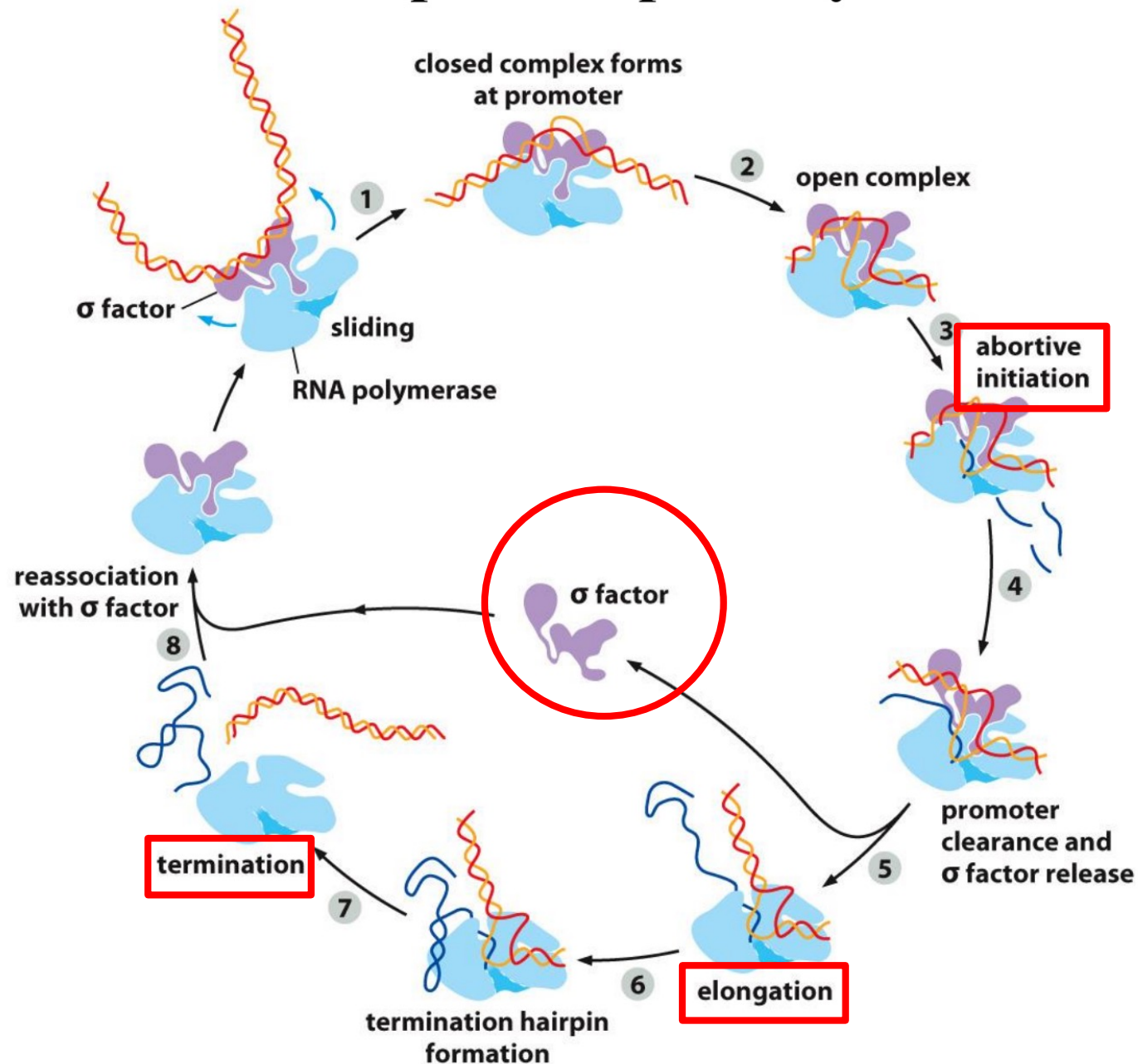
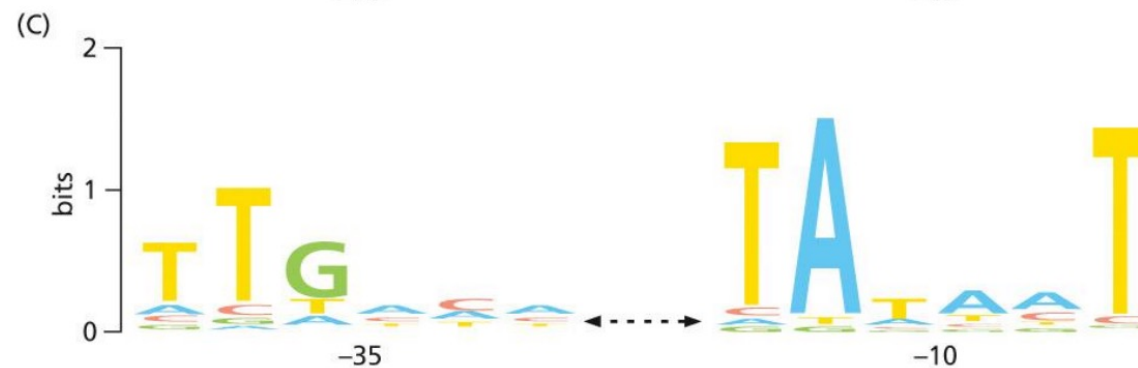
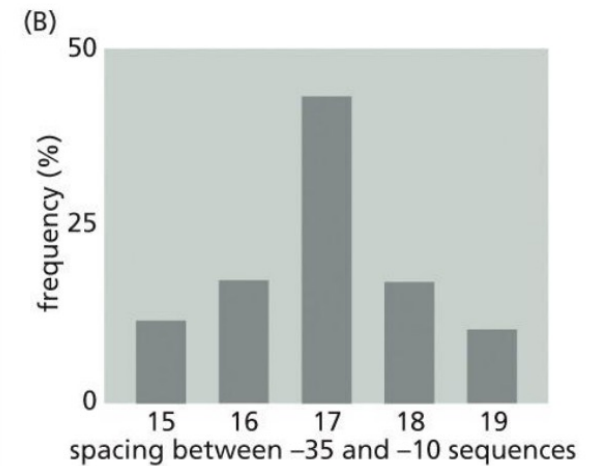
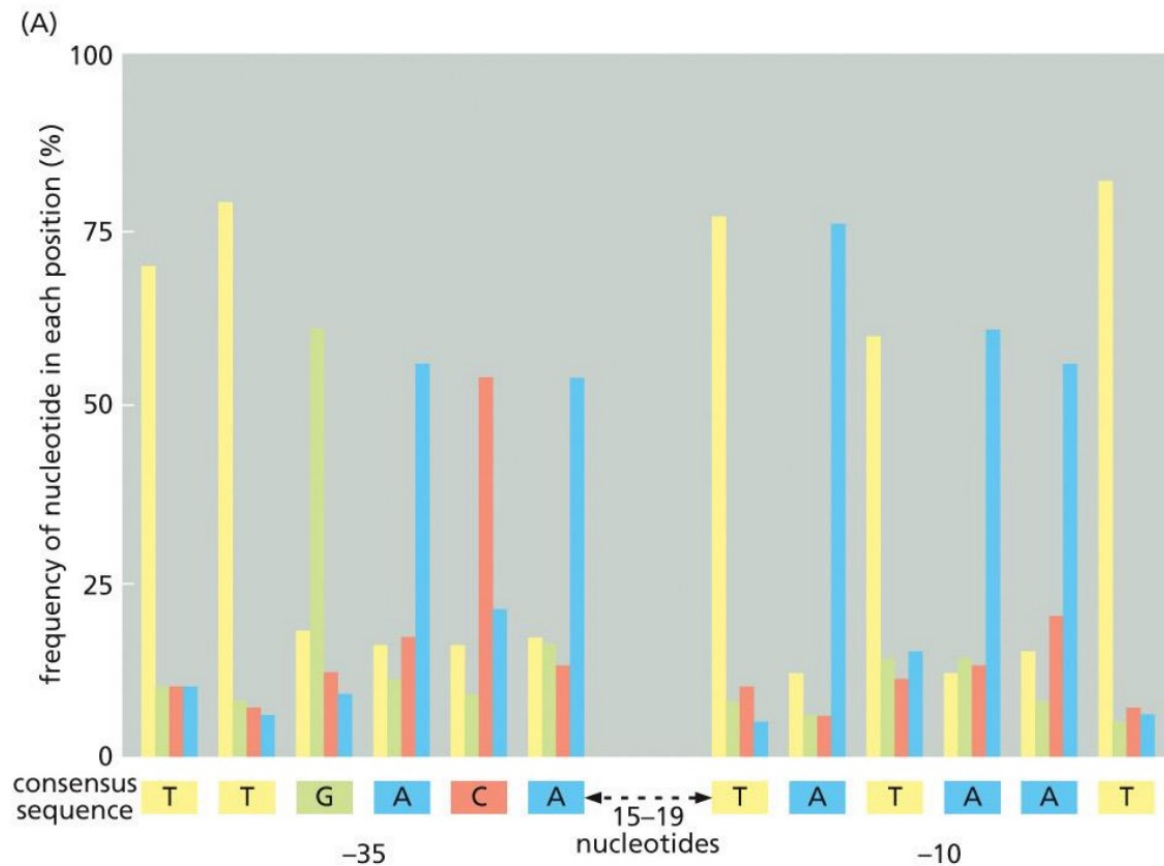


Figure 6-11a Molecular Biology of the Cell 6e (© Garland Science 2015)

Consensus binding sequence for σ factor



Asymmetry:
One orientation

Transcription in eukaryotes

- **Prokaryotes:** 1 RNA polymerase
- **Eukaryotes** (except plantae): 3 main RNA polymerases (RNA Pol I, II, and III)

Eukaryotes

1. **RNA polymerase I (Pol I):** Transcribes **rRNA** (specifically 28S, 18S, and 5.8S ribosomal RNA), which is crucial for ribosome structure and function.
2. **RNA polymerase II (Pol II):** Transcribes **mRNA**, which carries the genetic information from DNA to be translated into proteins. It also transcribes some small nuclear RNAs (**snRNA**), microRNAs (**miRNA**), long non-coding RNAs (**lncRNA**).
3. **RNA polymerase III (Pol III):** Transcribes **tRNA**, **5S rRNA**, and other small RNAs like **snRNA**.

Transcription in eukaryotes

- **Prokaryotes:** 1 RNA polymerase
- **Eukaryotes** (except plantae): 3 main RNA polymerases (RNA Pol I, II, and III)

Eukaryotes

TABLE 6–2 The Three RNA Polymerases in Eukaryotic Cells

Type of polymerase	Genes transcribed
RNA polymerase I	5.8S, 18S, and 28S rRNA genes
RNA polymerase II	All protein-coding genes, plus snoRNA genes, miRNA genes, siRNA genes, lncRNA genes, and most snRNA genes
RNA polymerase III	tRNA genes, 5S rRNA genes, some snRNA genes, and genes for other small RNAs
The rRNAs were named according to their “S” values, which refer to their rate of sedimentation in an ultracentrifuge. The larger the S value, the larger the rRNA.	

Transcription in eukaryotes

Transcription initiation

Prokaryotes

σ factor

Transcription in eukaryotes

Transcription initiation

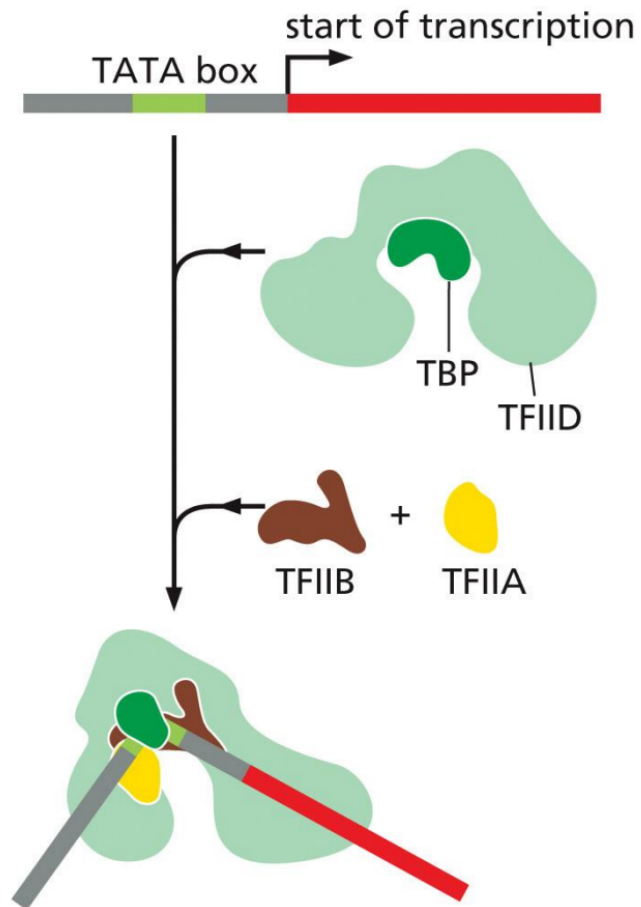
Eukaryotes

General Transcription
factors

TFIID, TFIIB, TFIIA, TFIIF, TFIIE, TFIIH...

Transcription in eukaryotes

Initiation

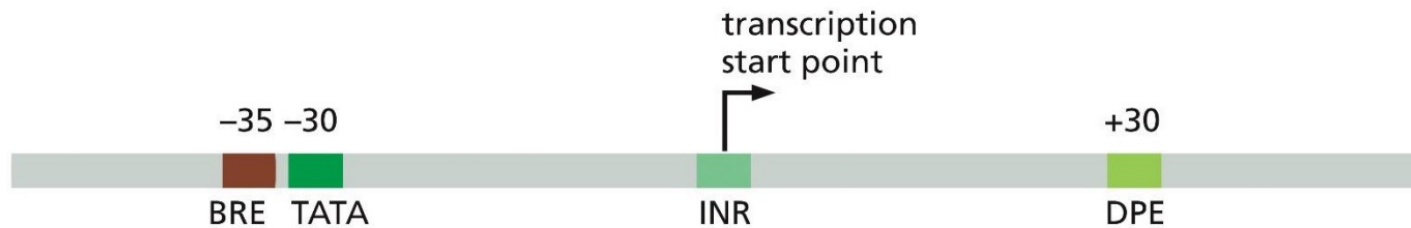


General Transcription factors:

TFIID:

TBP subunit (1 protein)

TAF subunits (~11 protein)



element	consensus sequence	general transcription factor
BRE	G/C G/C G/A C G C C	TFIIB
TATA	T A T A A/T A A/T	TBP subunit of TFIID
INR	C/T C/T A N T/A C/T C/T	TFIID
DPE	A/G G A/T C G T G	TFIID

BRE: TFIIB recognition element

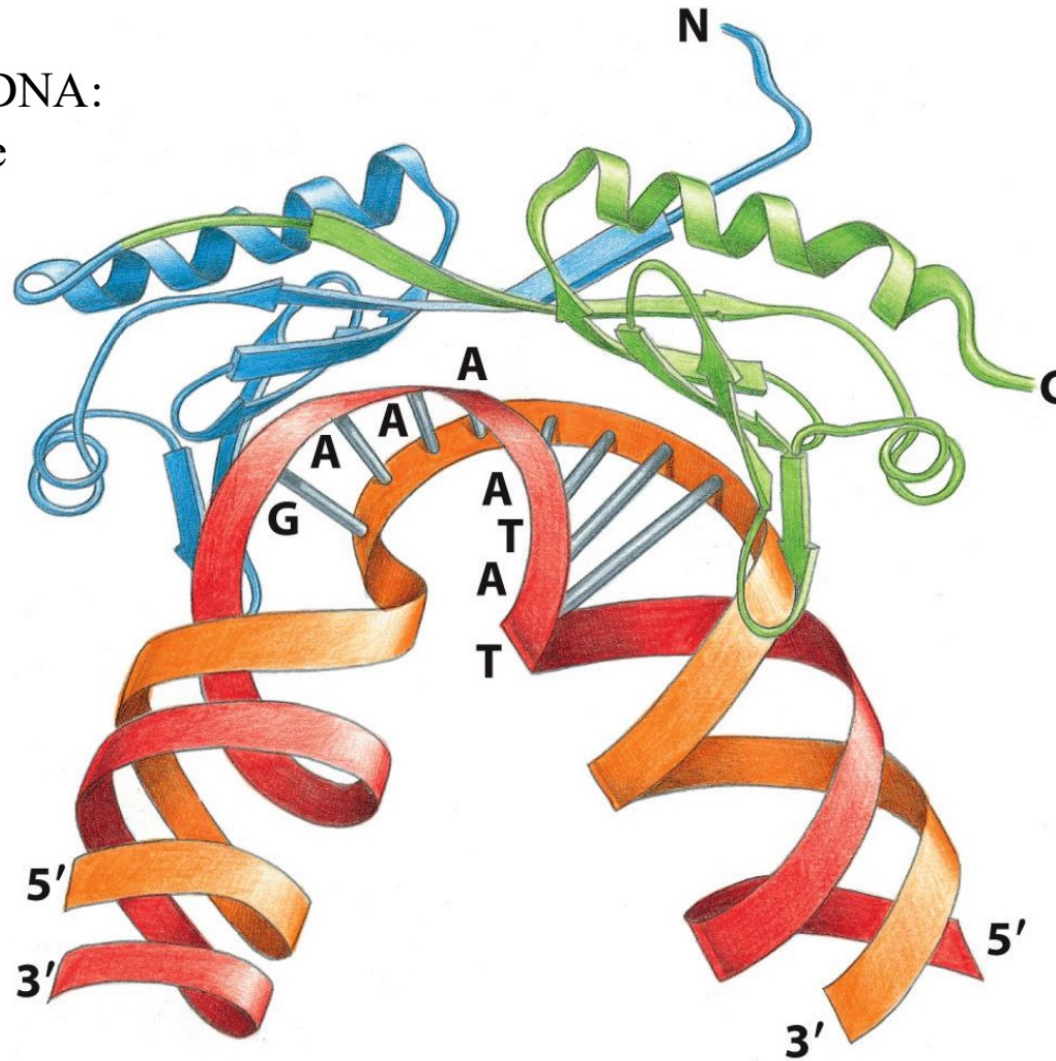
INR: initiator

DPE: downstream promoter element

Transcription in eukaryotes

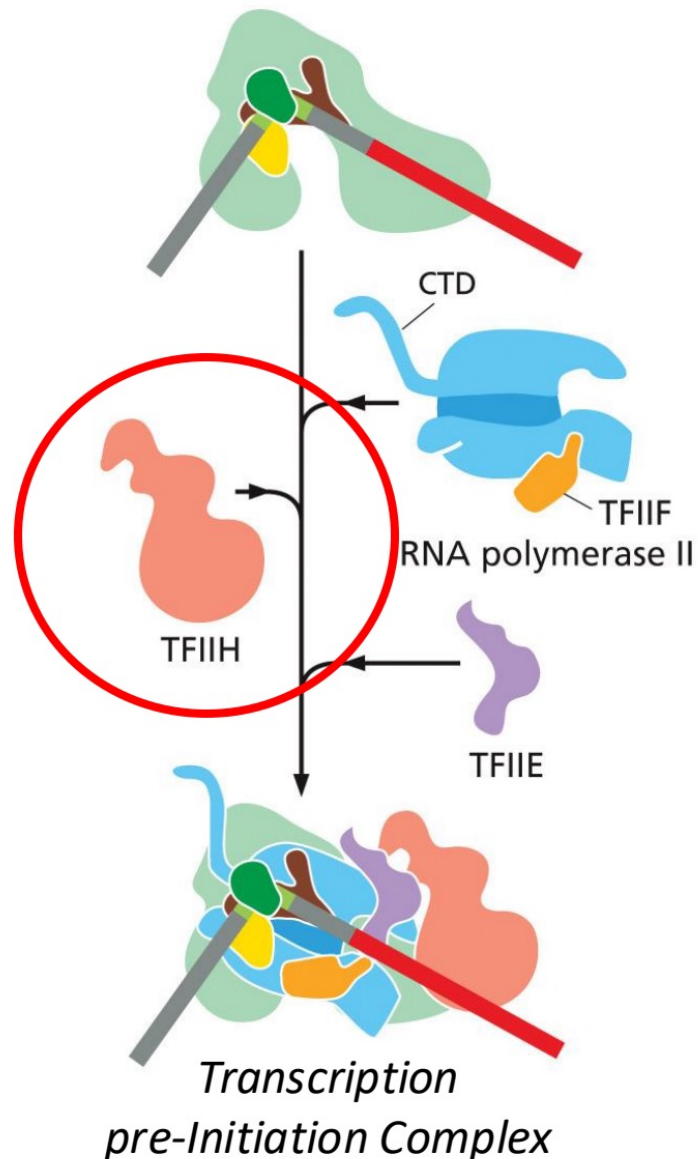
Initiation

Binding of TBP to DNA:
saddle-like structure



Transcription in eukaryotes

Initiation



CTD of RNA pol II:

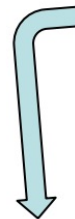
C-terminal domain of RNA pol II, carries enzymes for RNA processing.

Consists of 52 tandem repeats of a seven AA sequence.

Carries enzymes for RNA processing

Modification of CTD:

- serine 5 (initiation) phosphorylation
- serine 2 (elongation) phosphorylation



TFIIH: kinase activity (among others)

Transcription in eukaryotes

Initiation

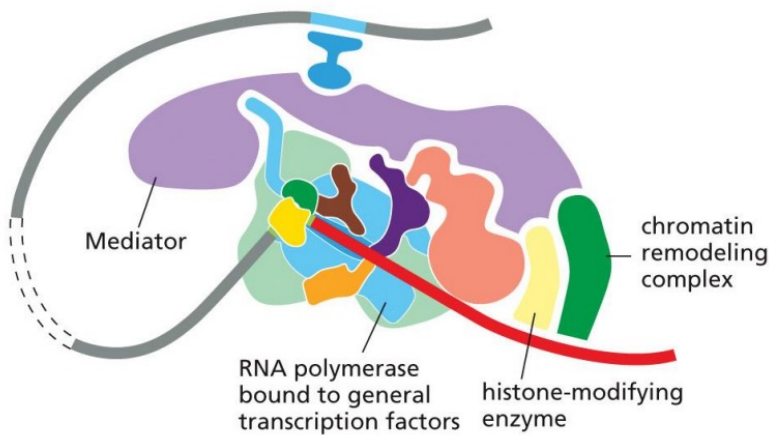


Transcription *pre-Initiation Complex*

Promoter recognition and assembly of the basic transcription machinery



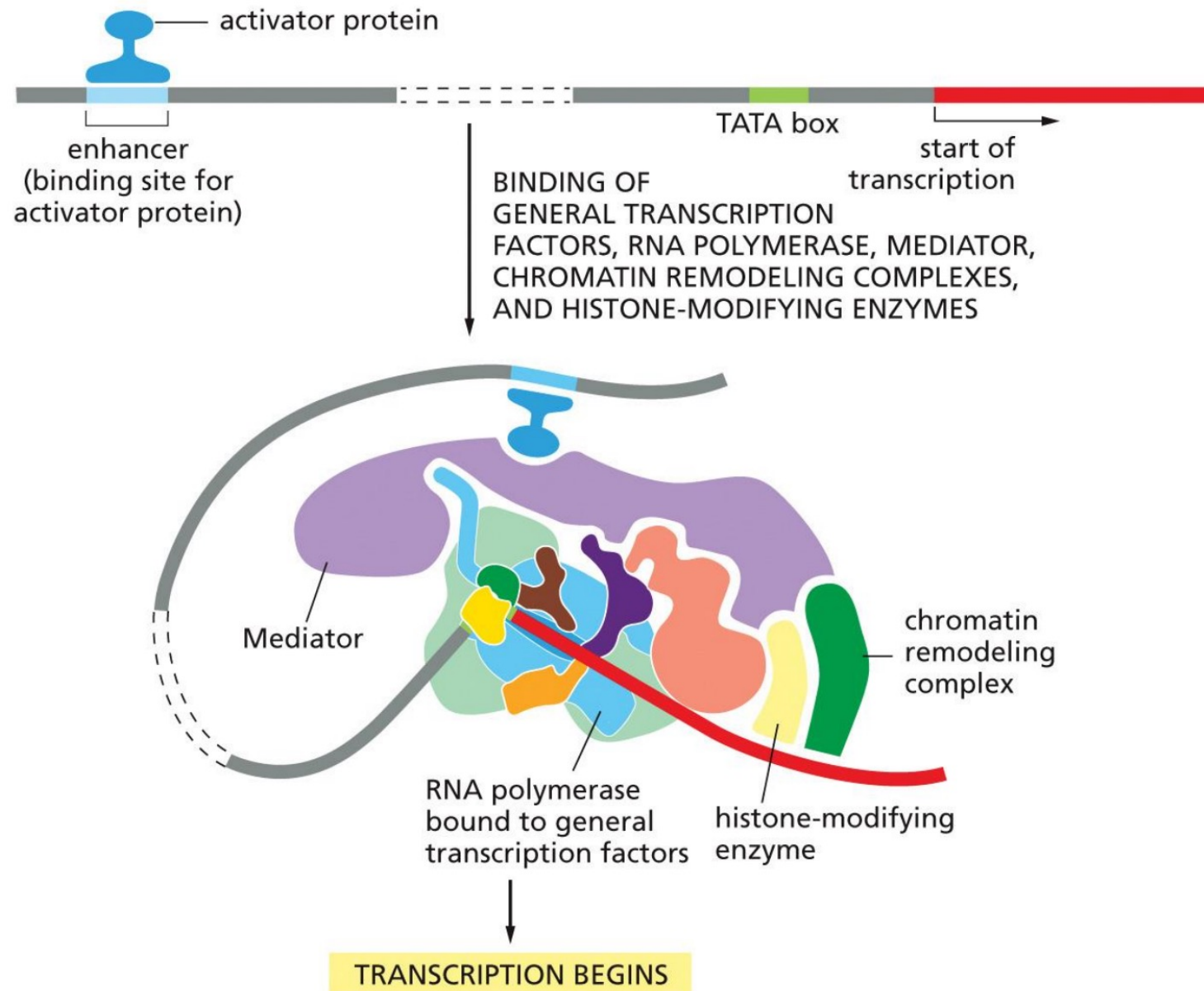
- Activators
- Mediators
- Chromatin Modifying enzymes (SAGA, ATAC)



Transcription *Initiation Complex*

RNA Pol II activation and start of transcription.

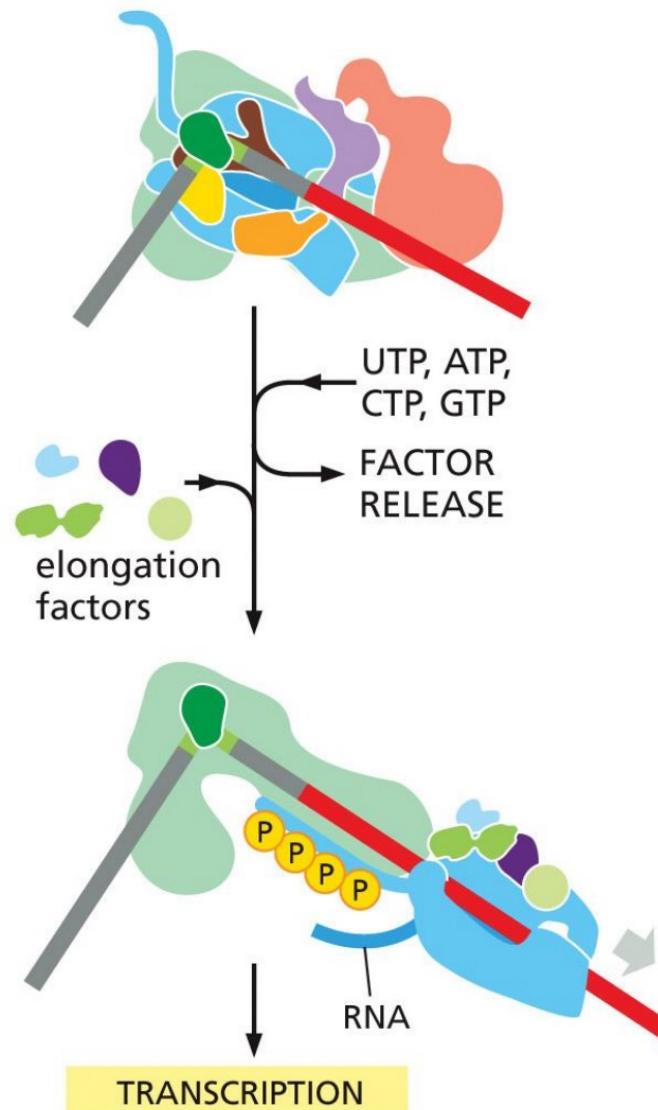
Looping of DNA to bring transcriptional regulators in close vicinity



**...what is the difference between a general transcription factor
and a sequence-specific transcription factor?**

Transcription in eukaryotes

Initiation to elongation



Initiation to elongation

Watch how promoter assembly, enhancer contact and RNA synthesis connect.



Transcription (Advanced) (1:53) — click the video to play

Cold Spring Harbor Laboratory, DNA Learning Center

...elongation and pre-mRNA processing.

Transcription in eukaryotes

- Shortly after initiation, both prokaryotic and eukaryotic **RNA polymerases** become **associated** with **elongation factors**, which decrease the likelihood of RNA polymerase dissociating from DNA (not shown in the video);
- In eukaryotes, **RNA polymerases** are associated with **ATP-dependent chromatin remodelling complexes**;
- **Transcription generates superhelical tension in the DNA.**

(A)

DNA with free end



unwind 10 DNA base pairs
(one helical turn)



DNA helix must
rotate one turn

(B)

DNA with fixed ends

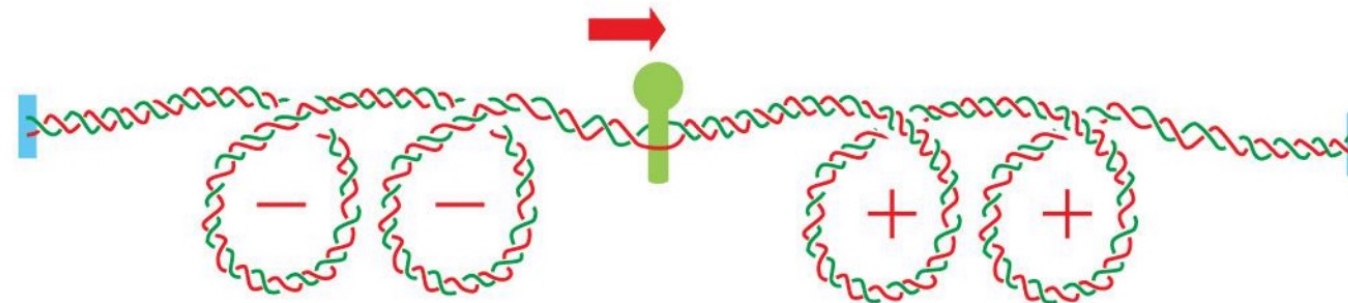
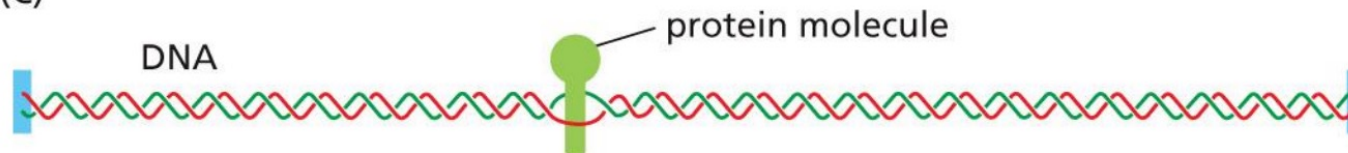


unwind 10 DNA base pairs
(one helical turn)



DNA helix forms
one supercoil

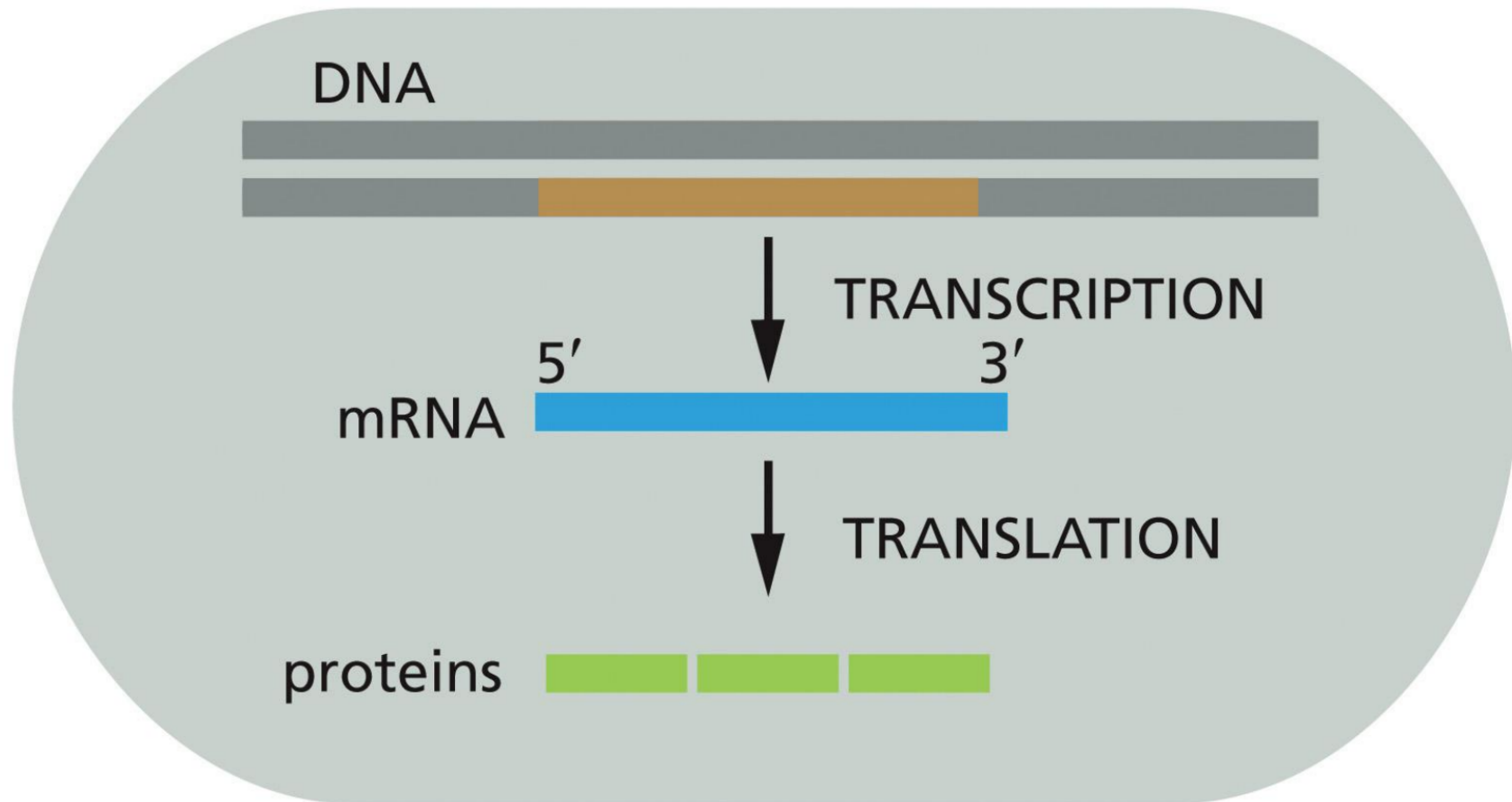
(C)



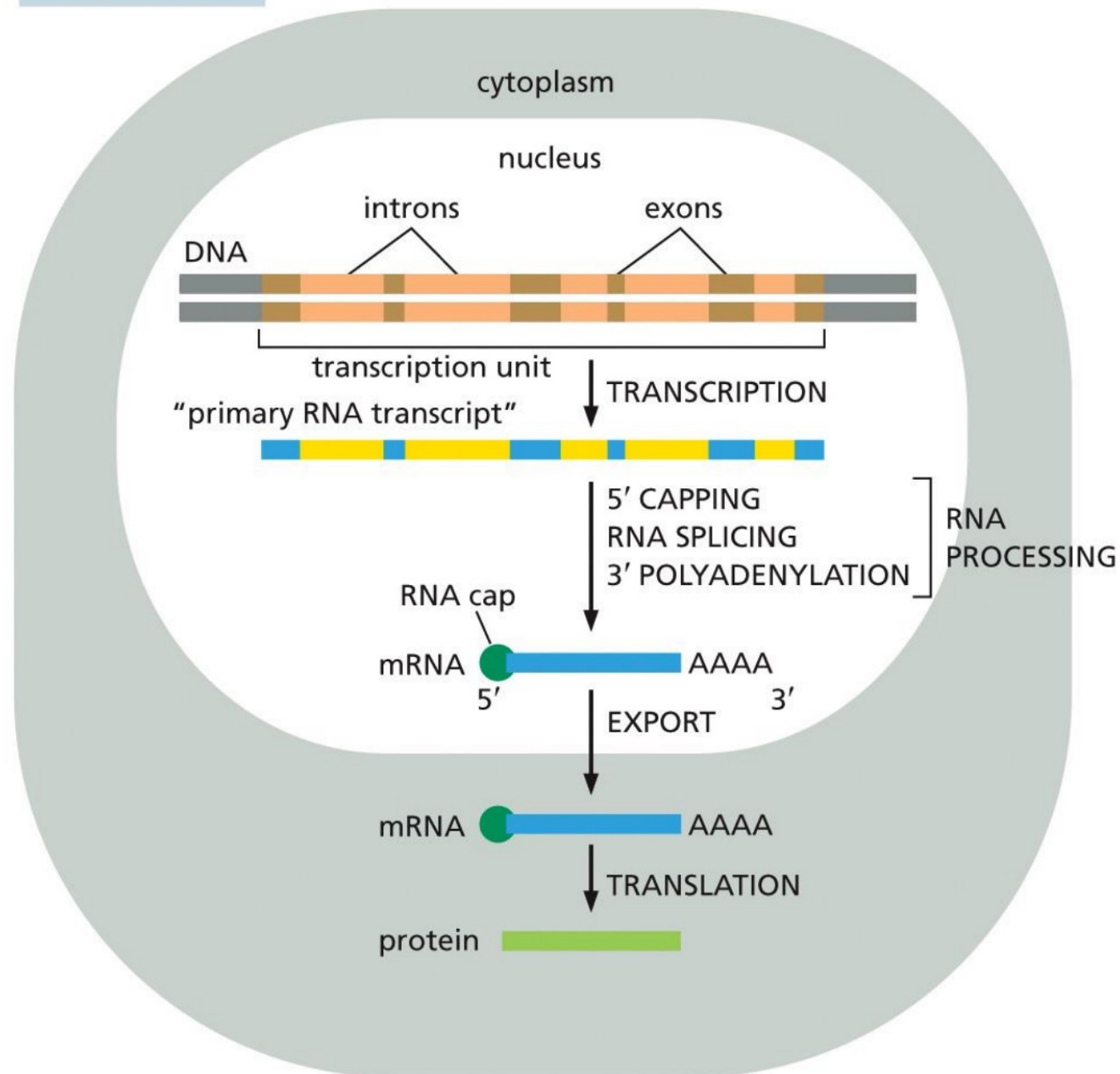
NEGATIVE SUPERCOILING
helix opening facilitated

POSITIVE SUPERCOILING
helix opening hindered

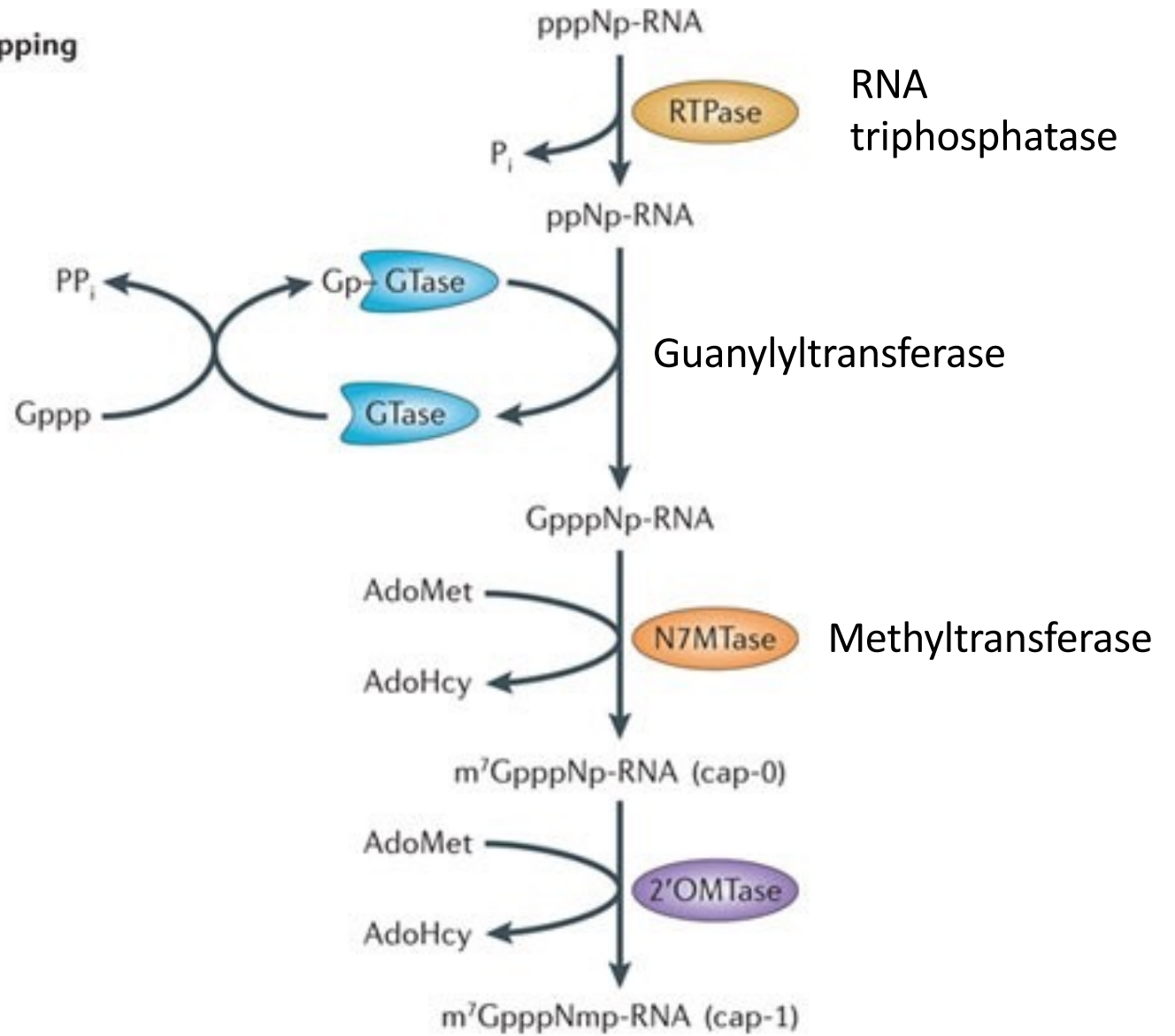
BACTERIA



EUKARYOTES



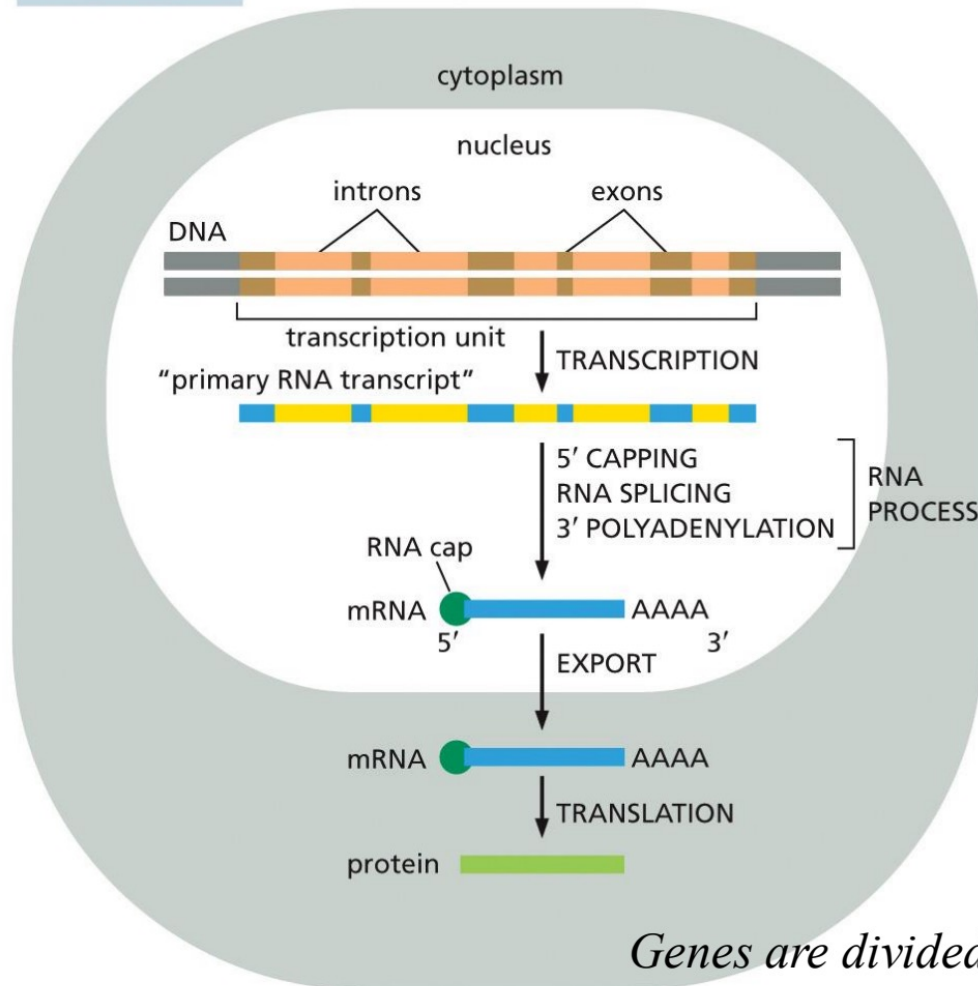
b Conventional RNA-capping pathway



...splicing.

mRNA Splicing

EUKARYOTES



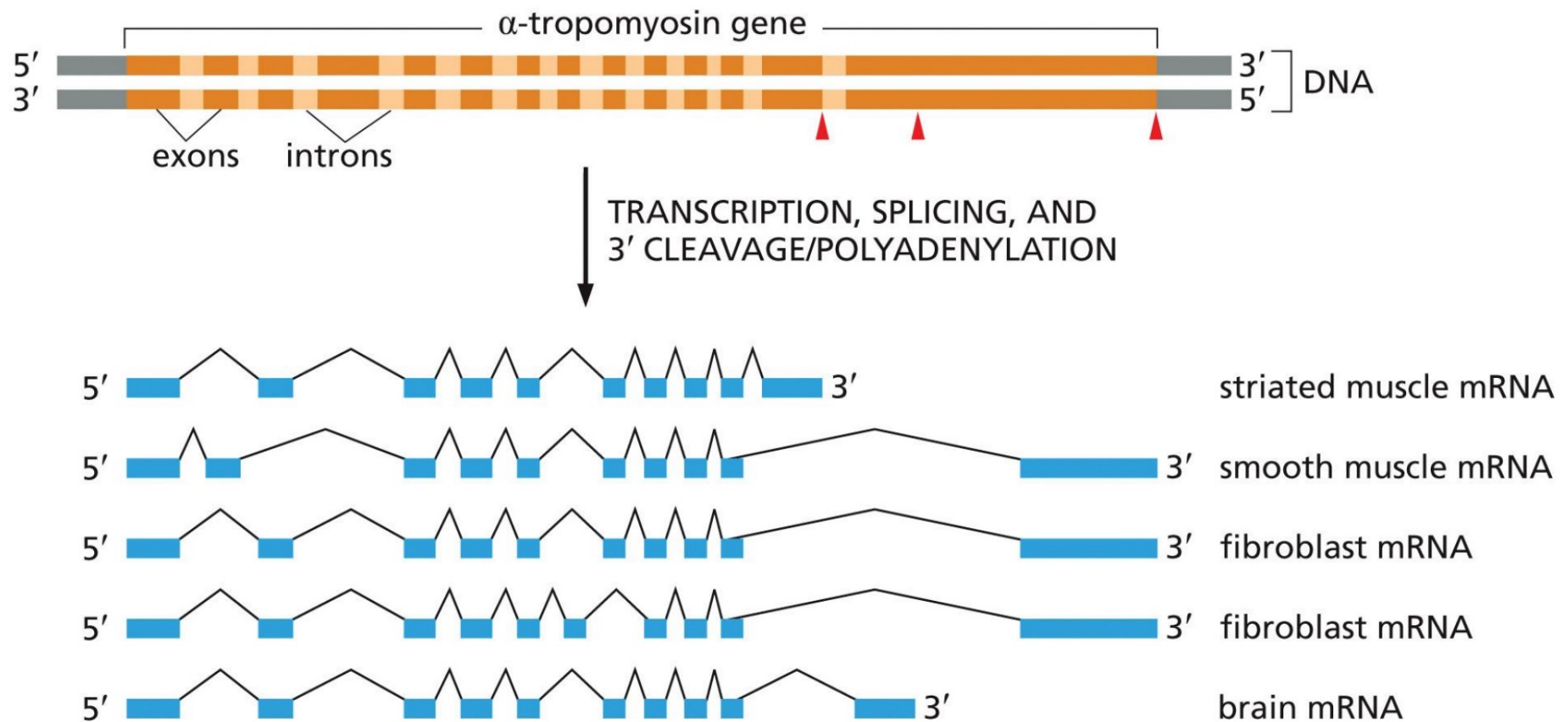
Splicing: pre-mRNA (but also rRNA, snRNA)

- Remove introns (10-10000 b) as a *lariat* via 2 sequential transesterifications reactions
- Reaction catalysed by several hundred proteins and 5 RNA molecules
- Energy: hydrolysis of (a lot of) ATP

WHY does it occur?

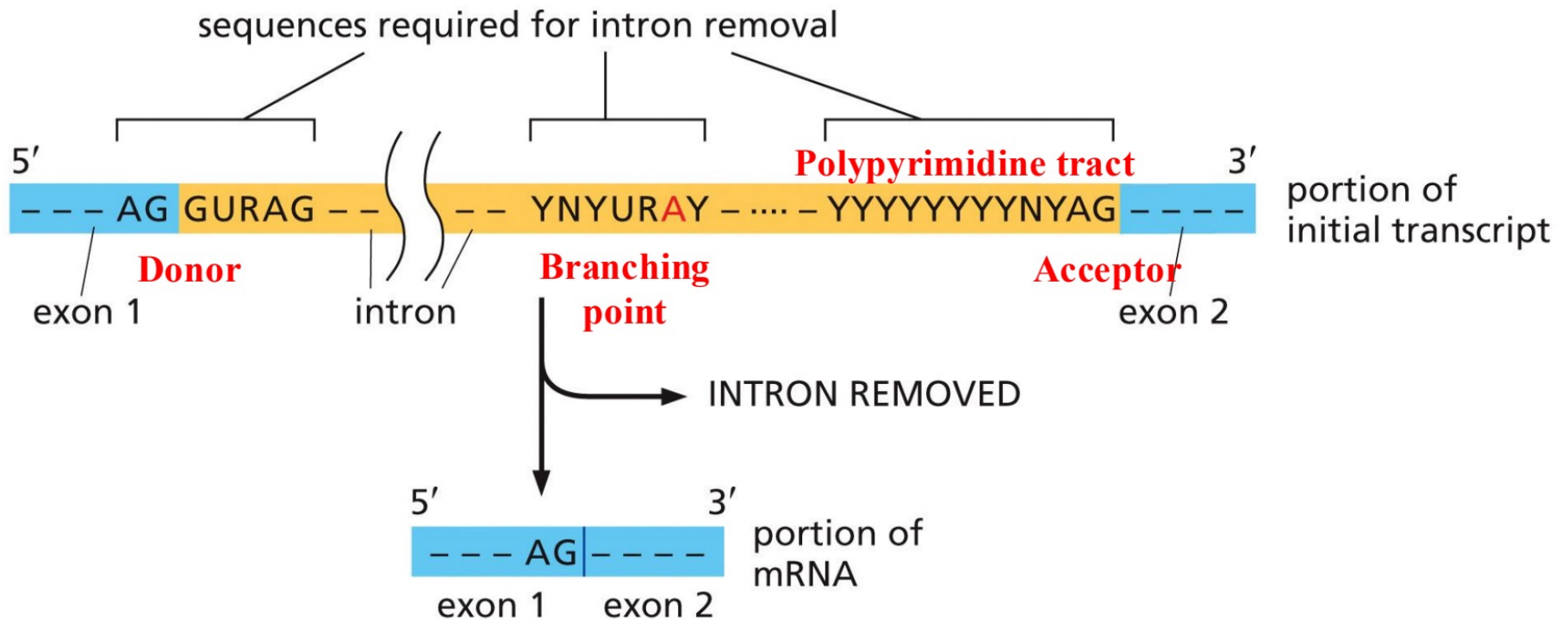
*Genes are divided in bits that can be combined in different ways to produce different proteins via a process called **ALTERNATIVE SPLICING** (evolutionary advantageous); This also ensure precision during splicing; also advantageous for the make up of new proteins (think about intron size)*

Example of alternative splicing

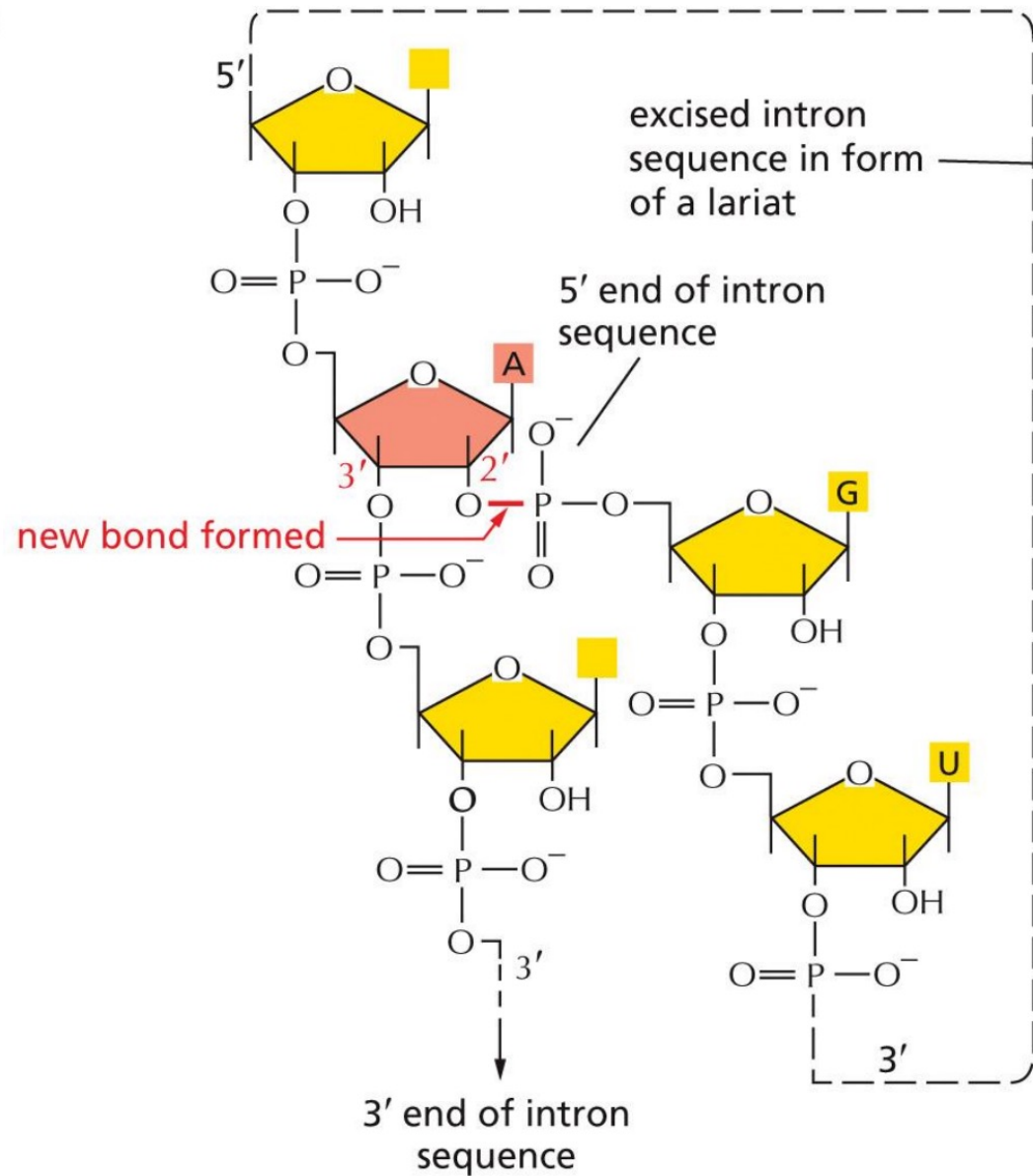
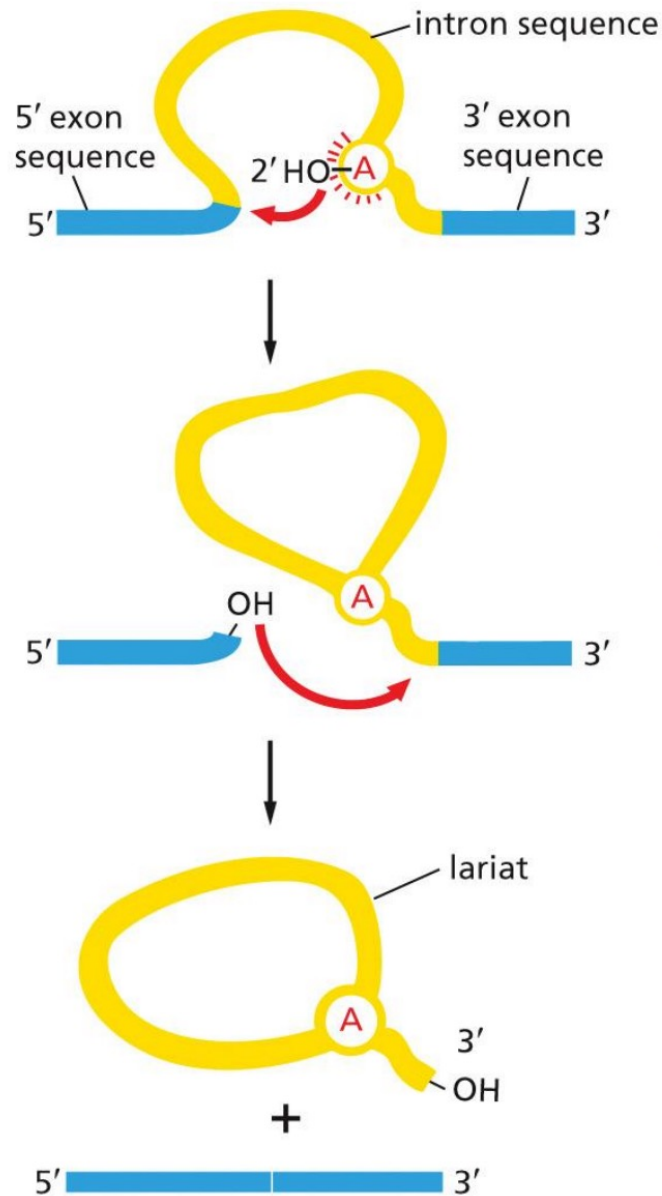


Donor site, acceptor site, and branch point sequences are short and can vary = alternative splicing

Splicing sequences

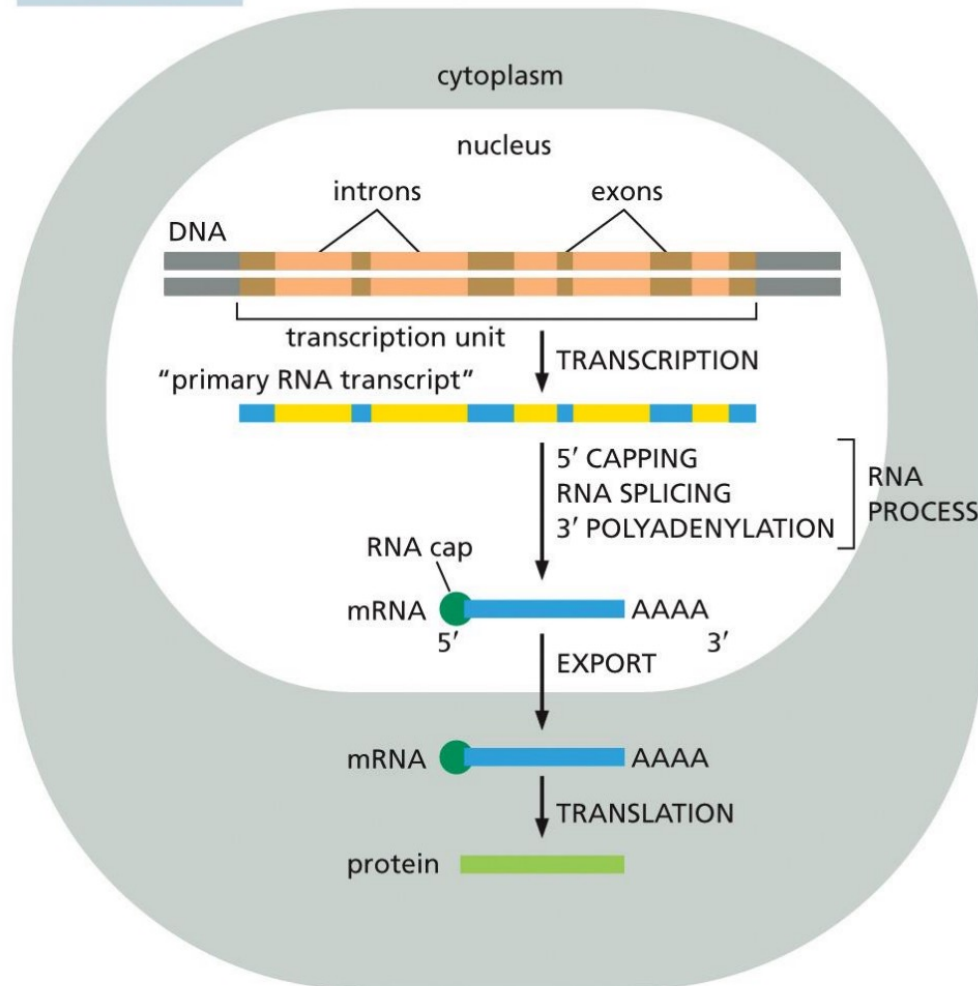


Splicing Reaction



mRNA Splicing

EUKARYOTES

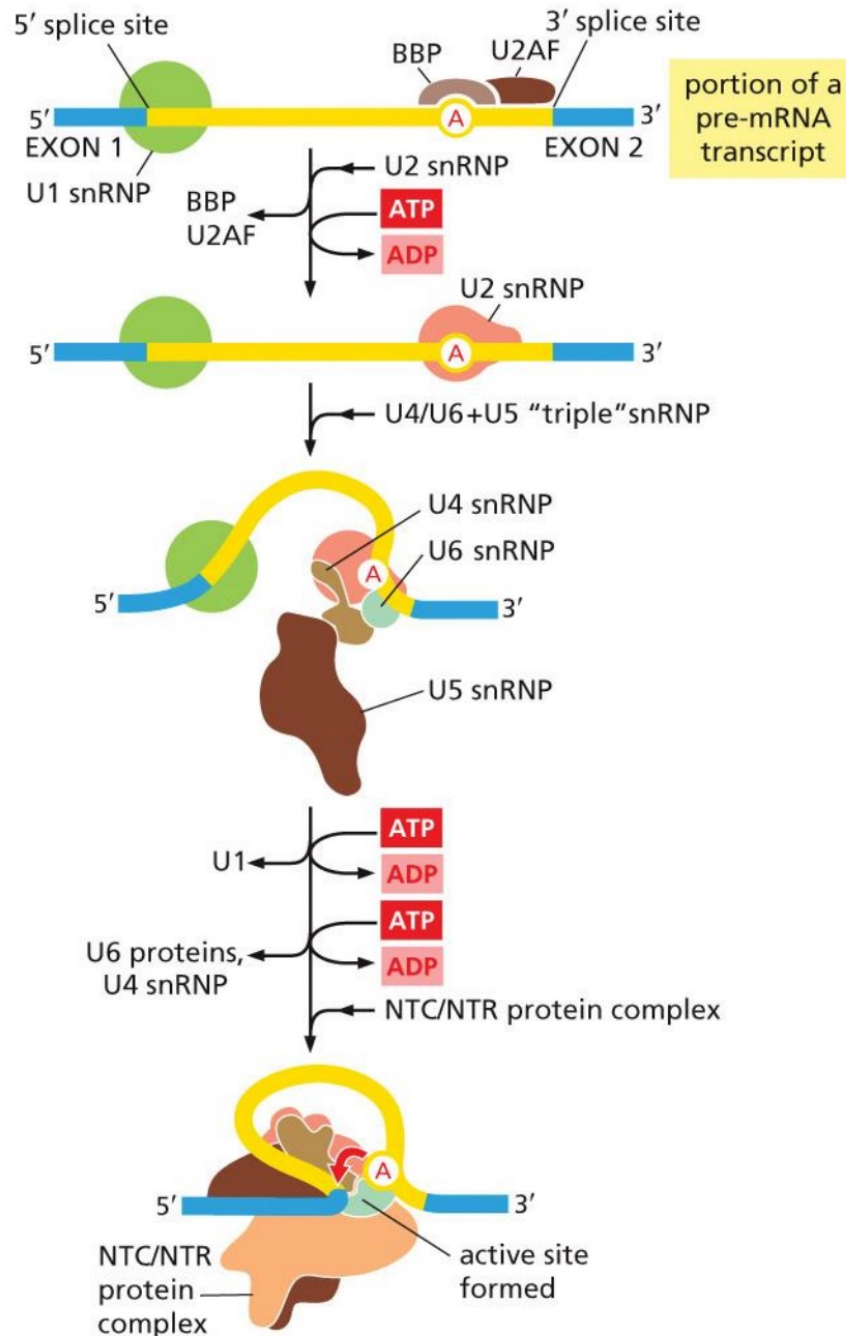


Splicing: pre-mRNA (but also rRNA, snRNA)

- Remove introns (10-10000 nt) as a *lariat* via 2 sequential transesterifications reactions
- Reaction catalysed by several hundred proteins and 5 RNA molecules

Small-nuclear RNAs: U1, U2, U4, U5, U6

- < 200 nt each
- Each complexed with at least 7 proteins to form a snRNP
- Together form the core spliceosome

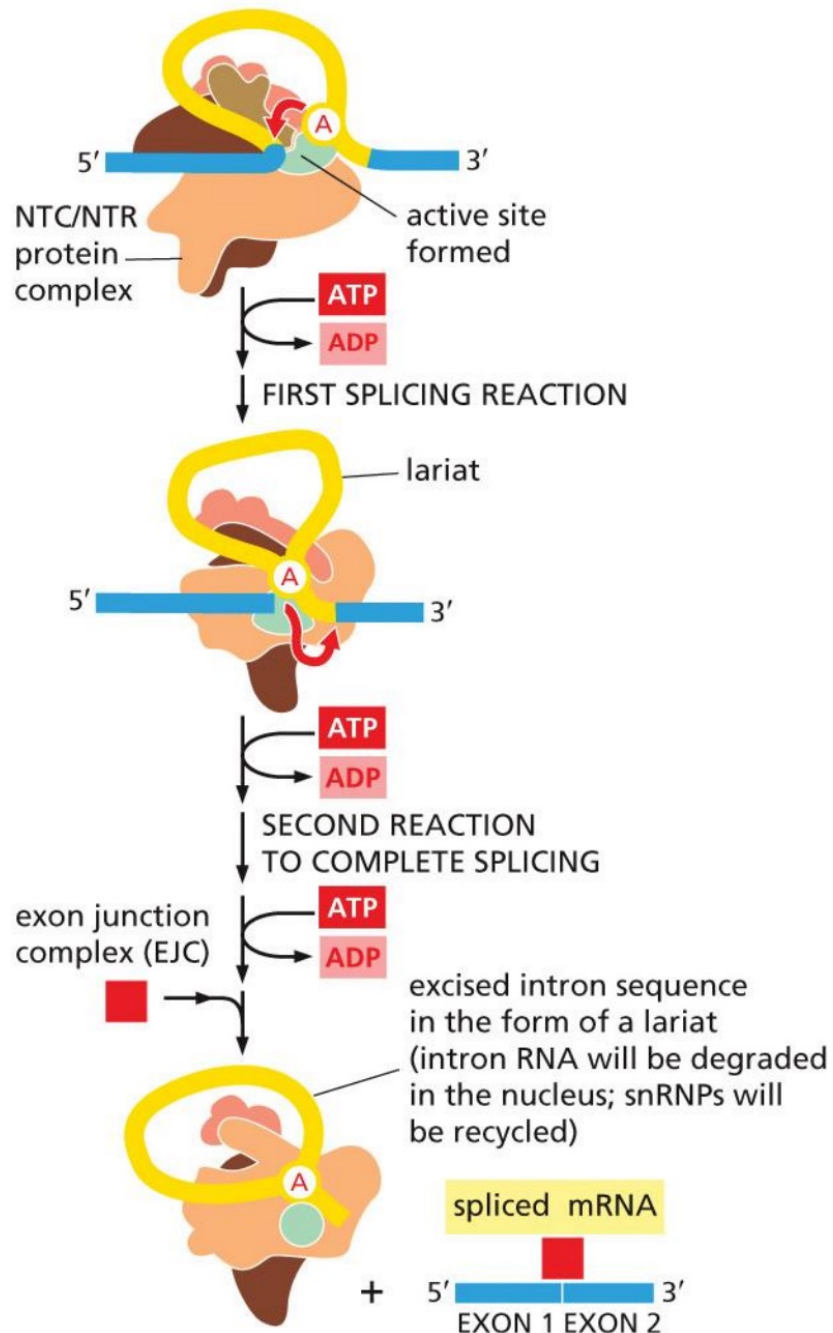


The U1 snRNP forms base pairs with the 5' splice junction. BBP (branch-point binding protein) recognizes the branch-point site and binds cooperatively with U2AF, which recognizes the polypyrimidine tract and the 3' splice junction (see Figure 6–28).

The U2 snRNP displaces BBP and U2AF and forms base pairs with the branch-point site consensus sequence.

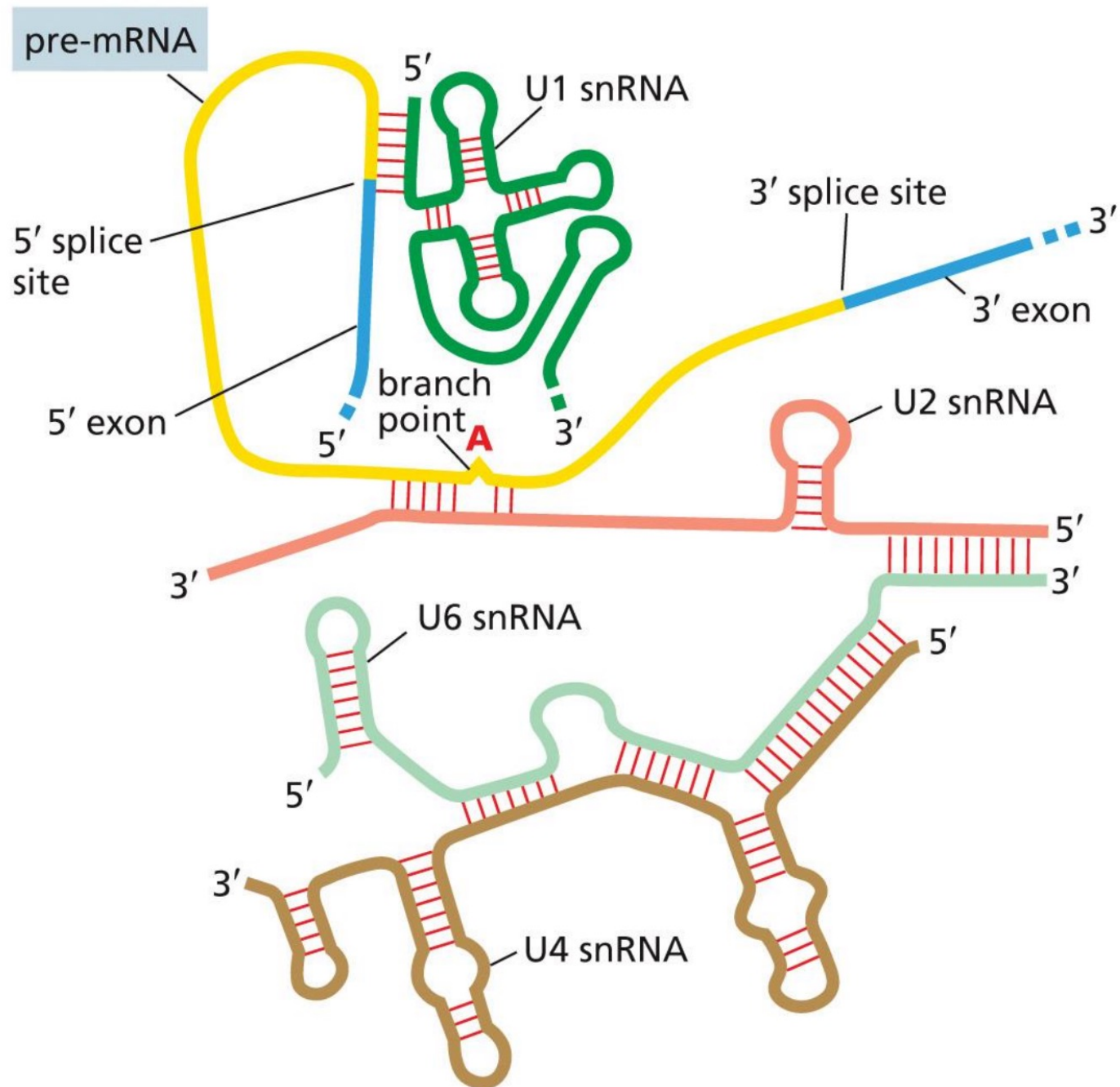
The U4/U6+U5 "triple" snRNP enters the reaction. In this triple snRNP, the U4 and U6 snRNAs are held firmly together by base-pair interactions. Subsequent rearrangements break apart the U4/U6 base pairs, allowing U6 to displace U1 at the 5' splice junction and ejecting the U4 snRNP and some of the proteins of the U6 snRNP.

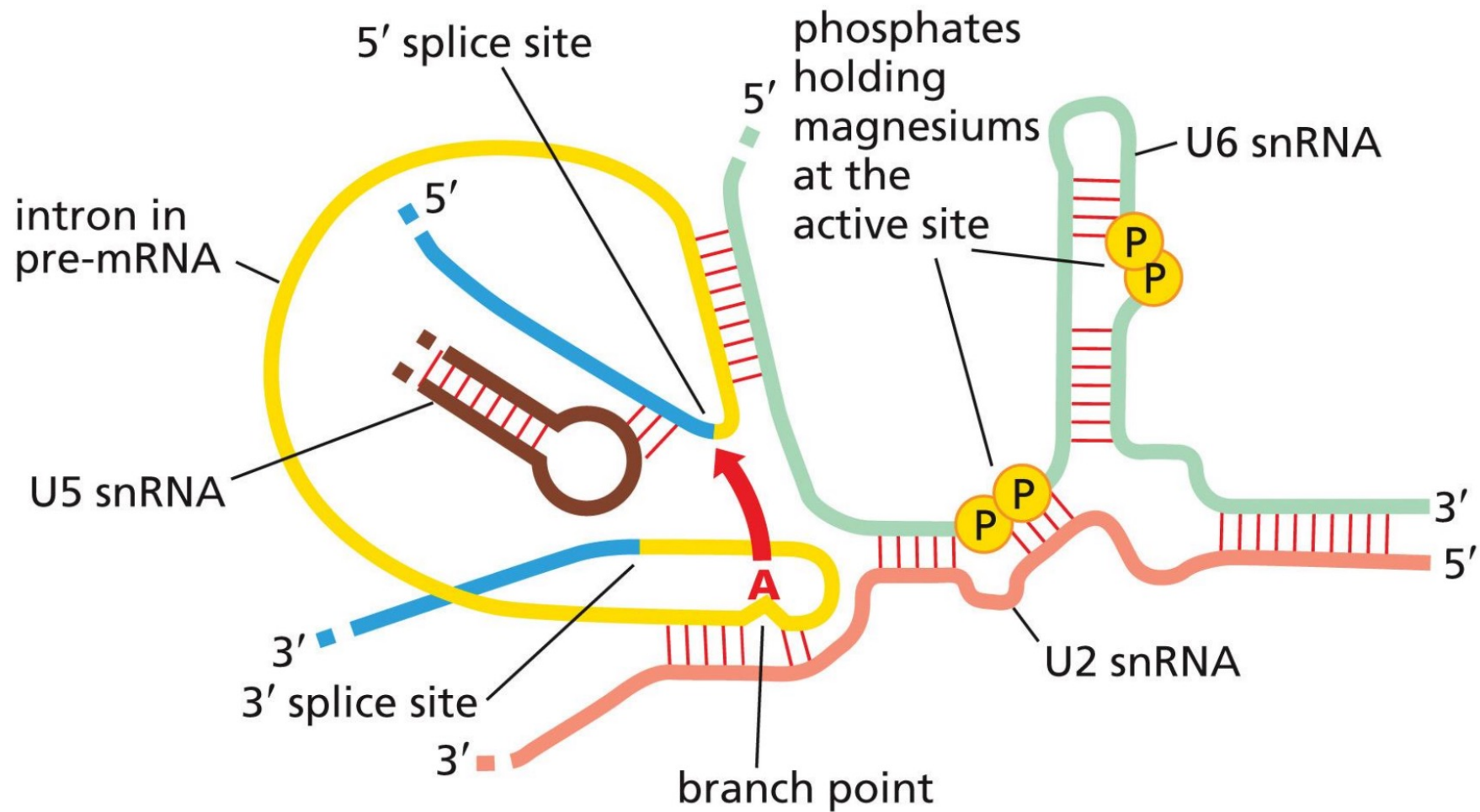
Addition of the NTC/NTR protein complex positions the snRNPs to form the active site of the spliceosome and brings the branch point in proximity to the 5' splice site.



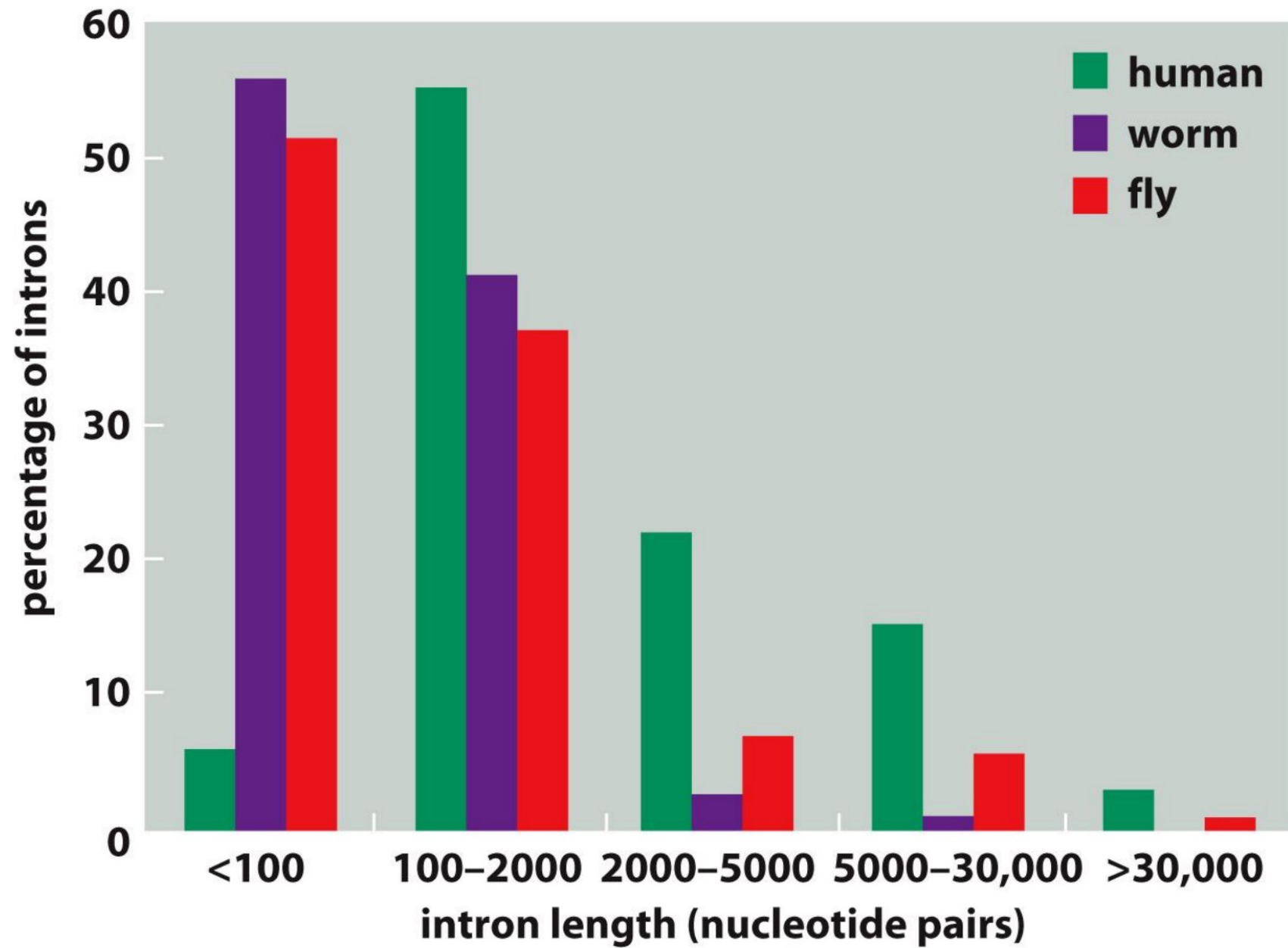
Lariat formed. Additional rearrangements bring the two exon segments together and place them in the active site.

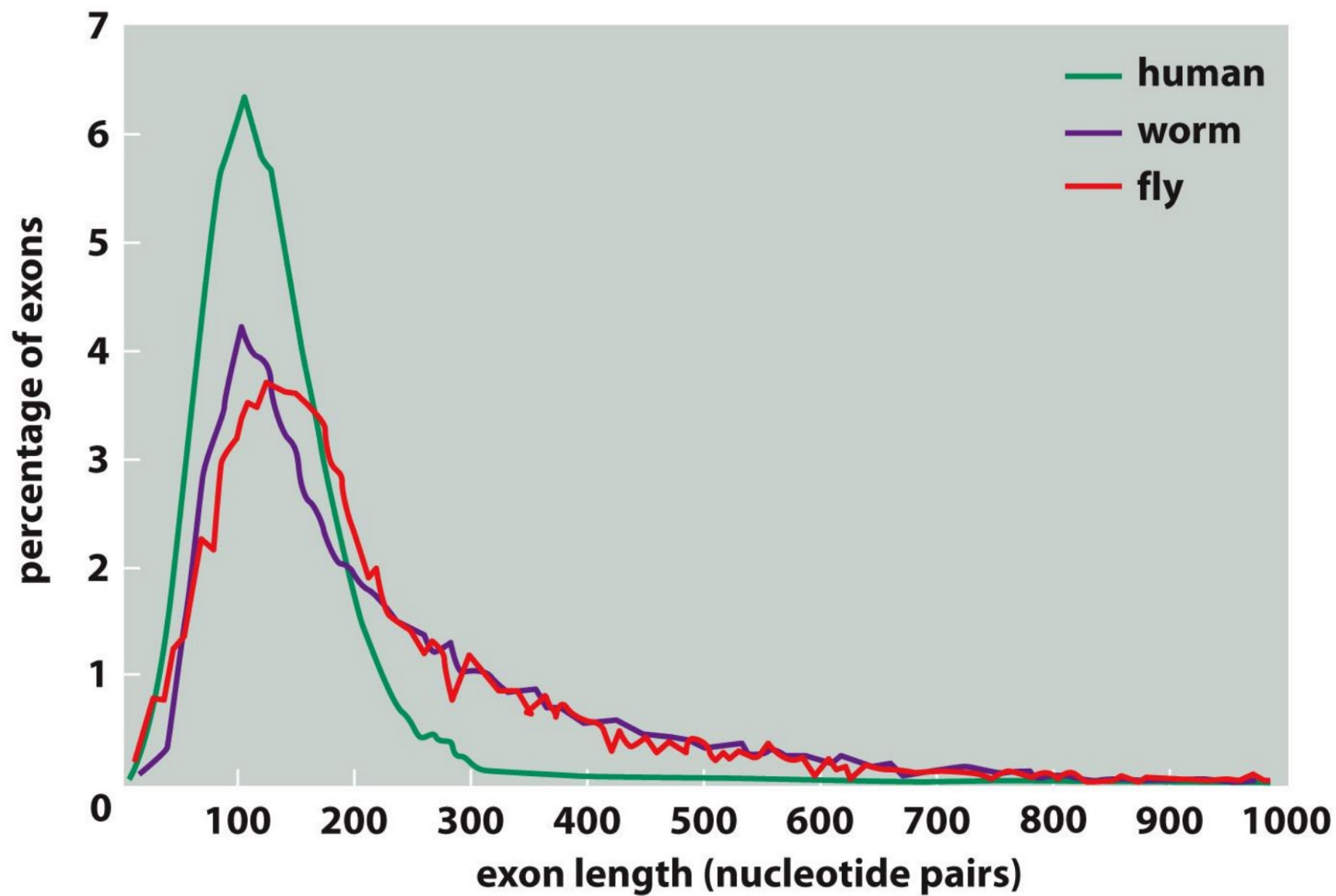
Spliced RNA is released. Spliceosome components recycled using ATP hydrolysis to "reset" them.

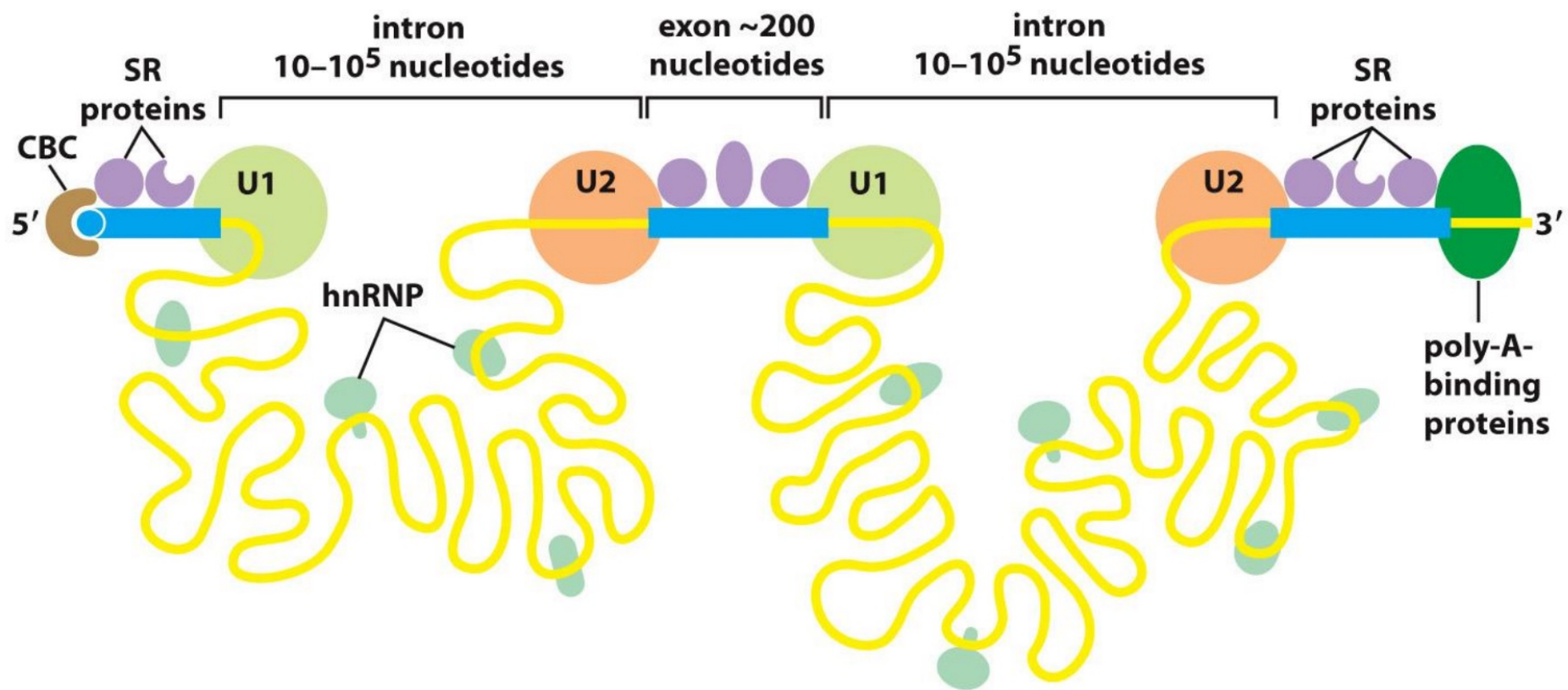




Adapted from M.E. Wilkinson, C. Charenton, and K. Nagai, *Annu. Rev. Biochem.* 89:359–398, 2020. With permission from Annual Reviews.







The spliceosome

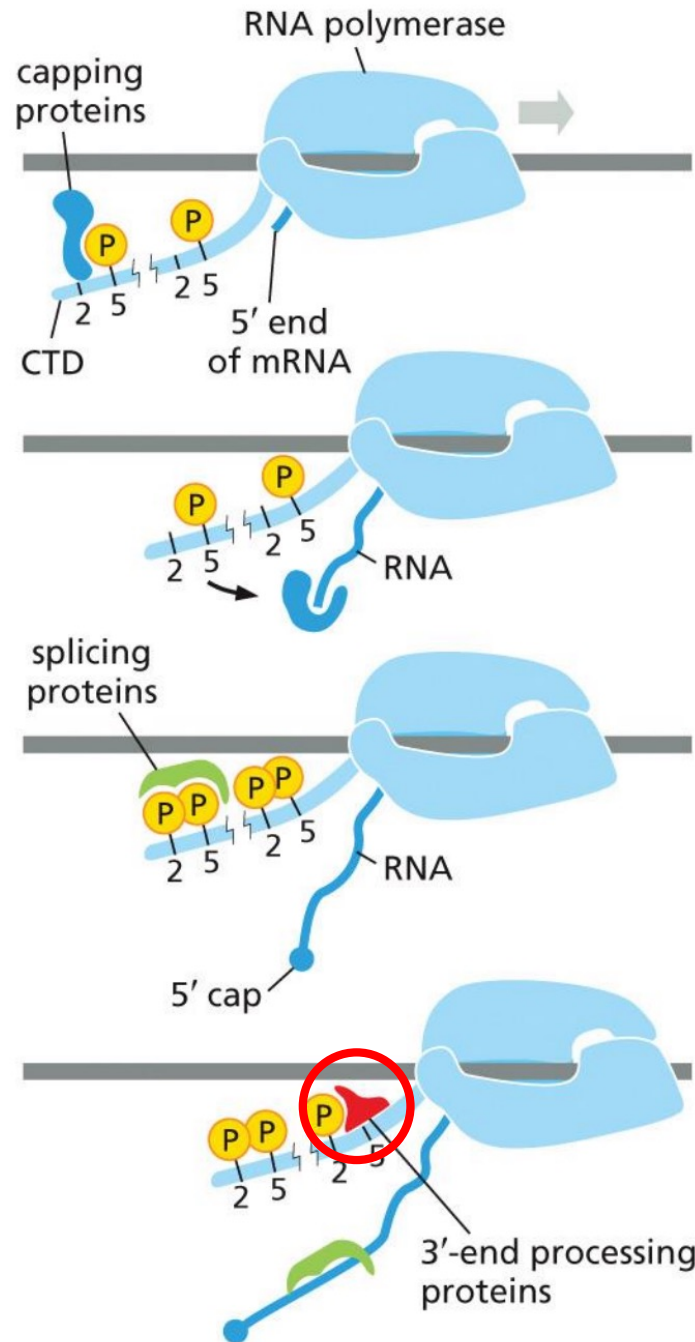
Follow spliceosome assembly, lariat formation and exon joining.



[RNA Splicing \(1:37\)](#) — click the video to play

Cold Spring Harbor Laboratory, DNA Learning Center

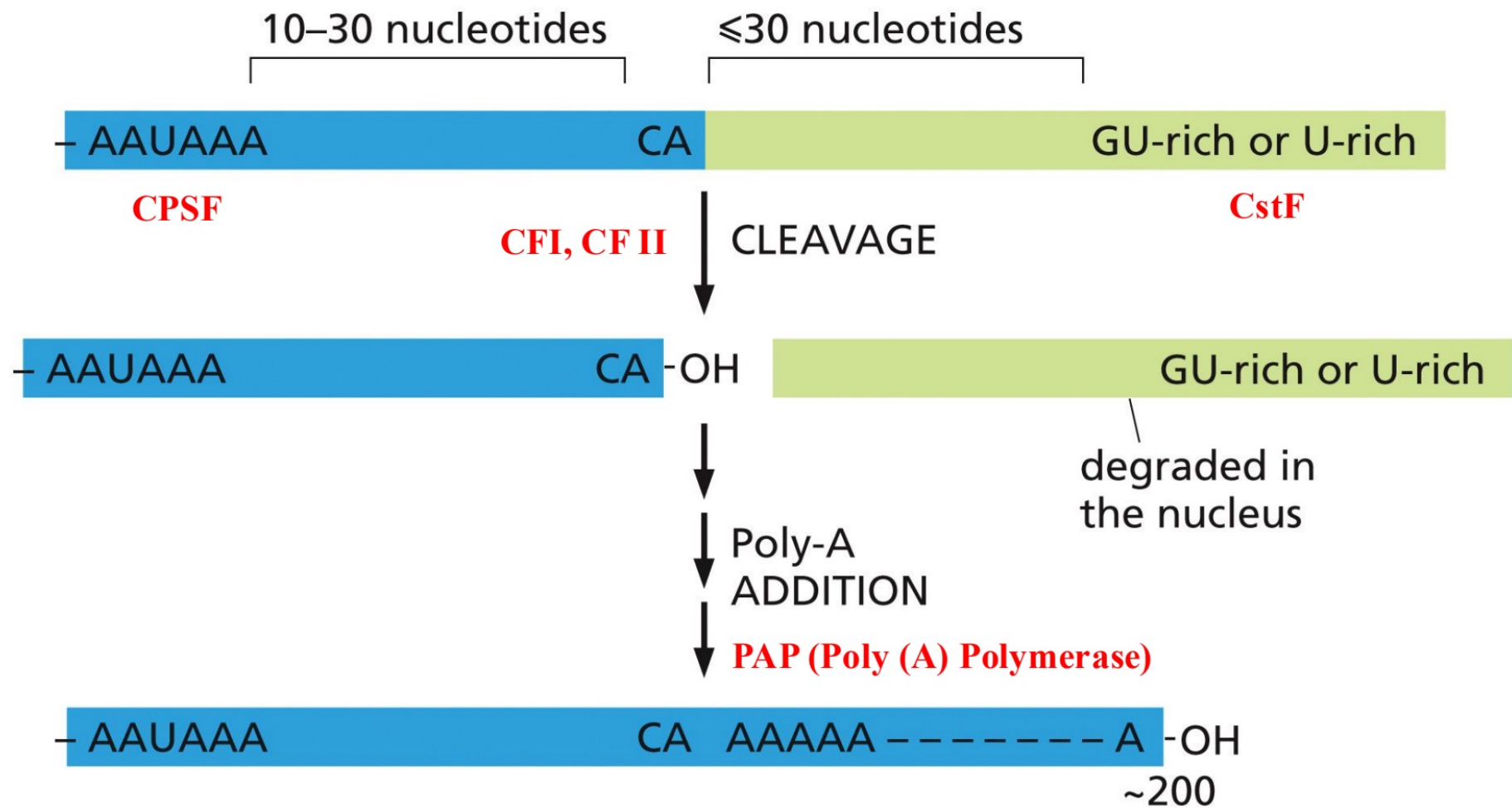
...termination.

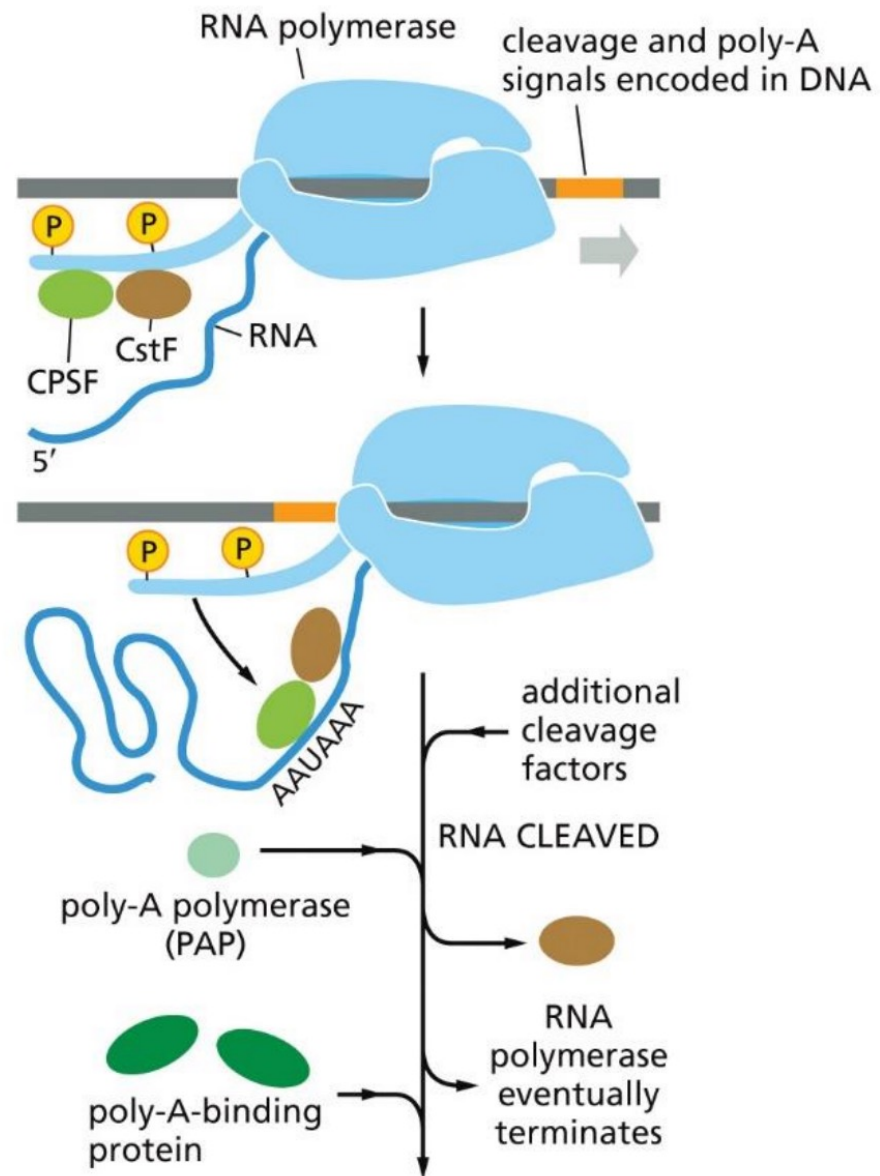


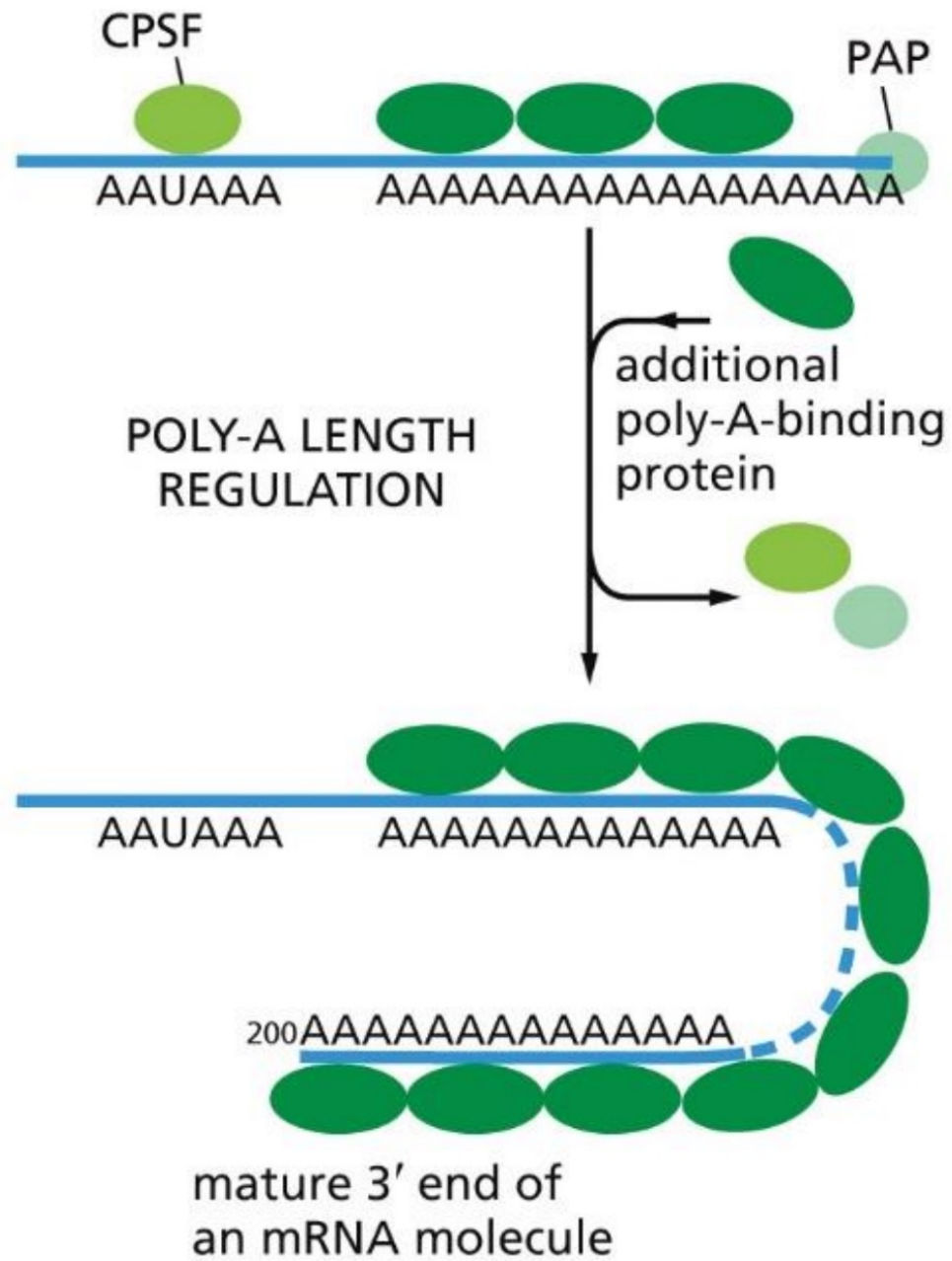
CPSF: *Cleavage and polyadenylation specificity factor*

CstF: *Cleavage stimulation factor*

Termination
sequence is specified
in the genome

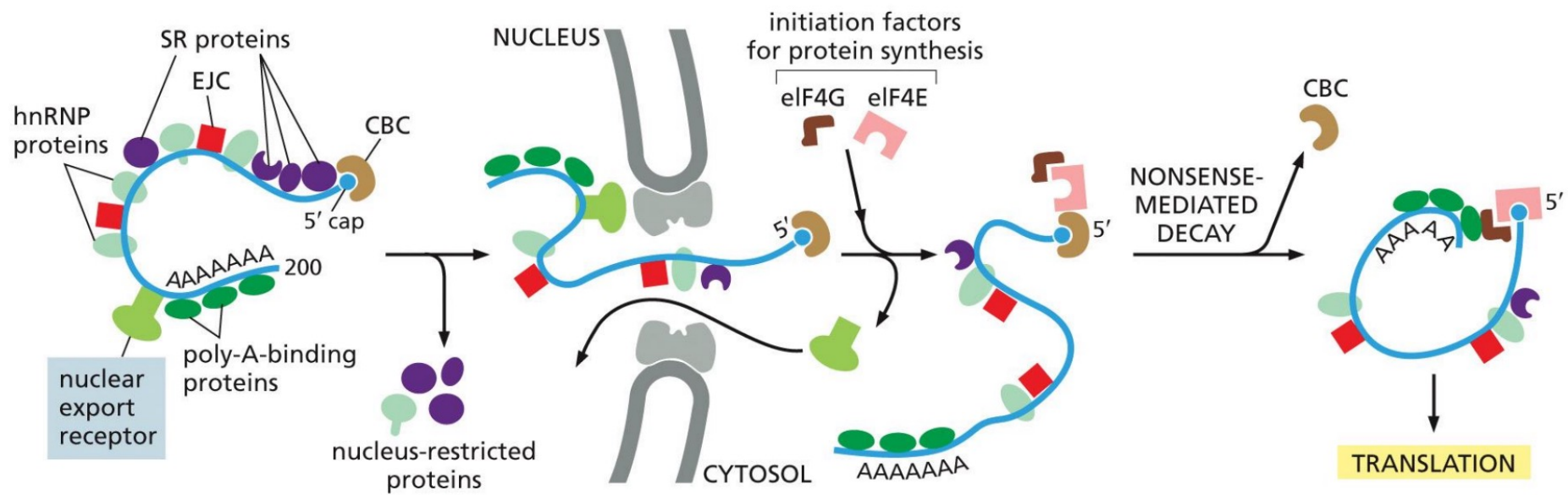


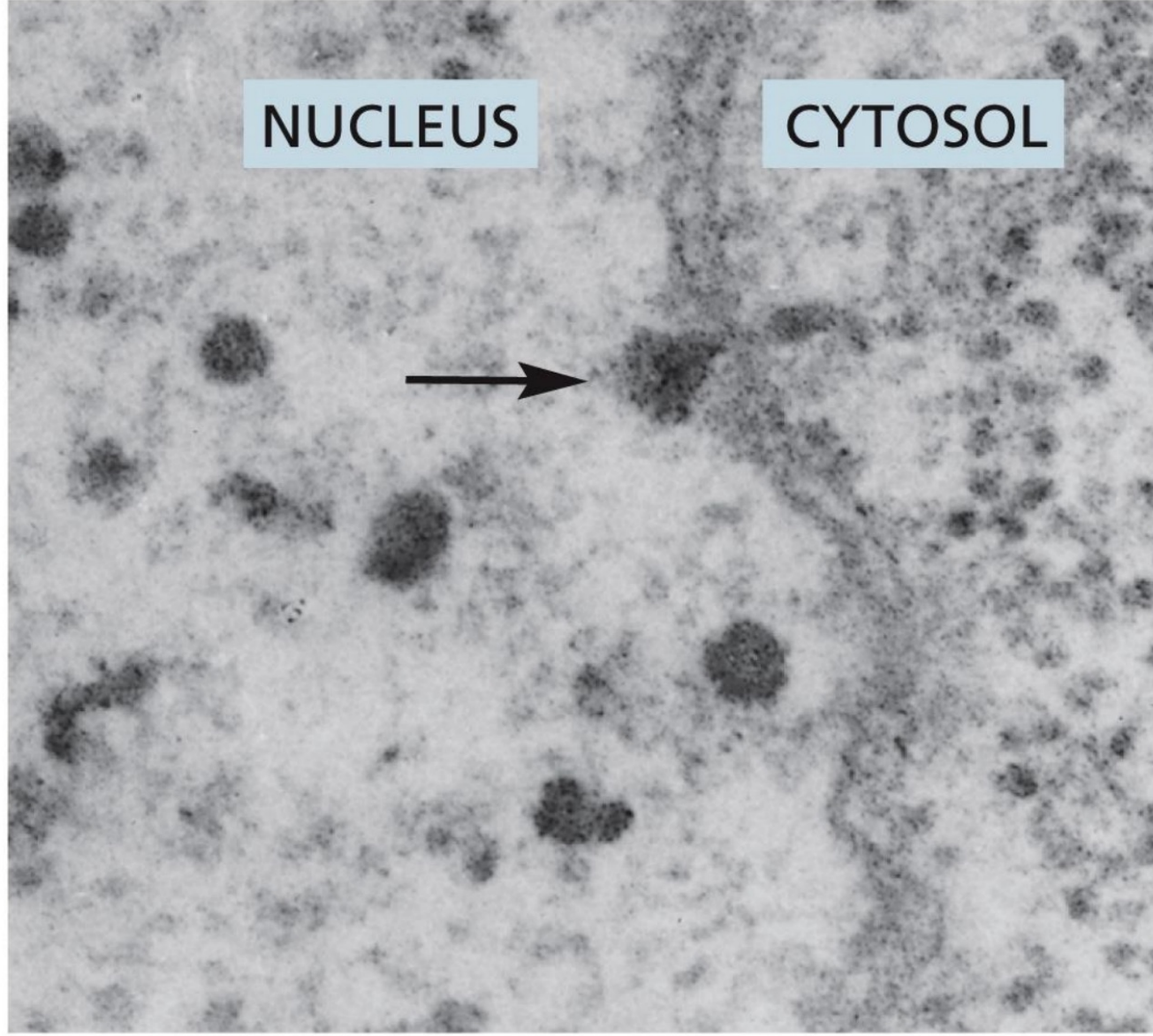




**...What happens with the mRNA after
transcription termination and splicing?**

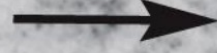
...And with the RNA polymerase?





CYTOSOL

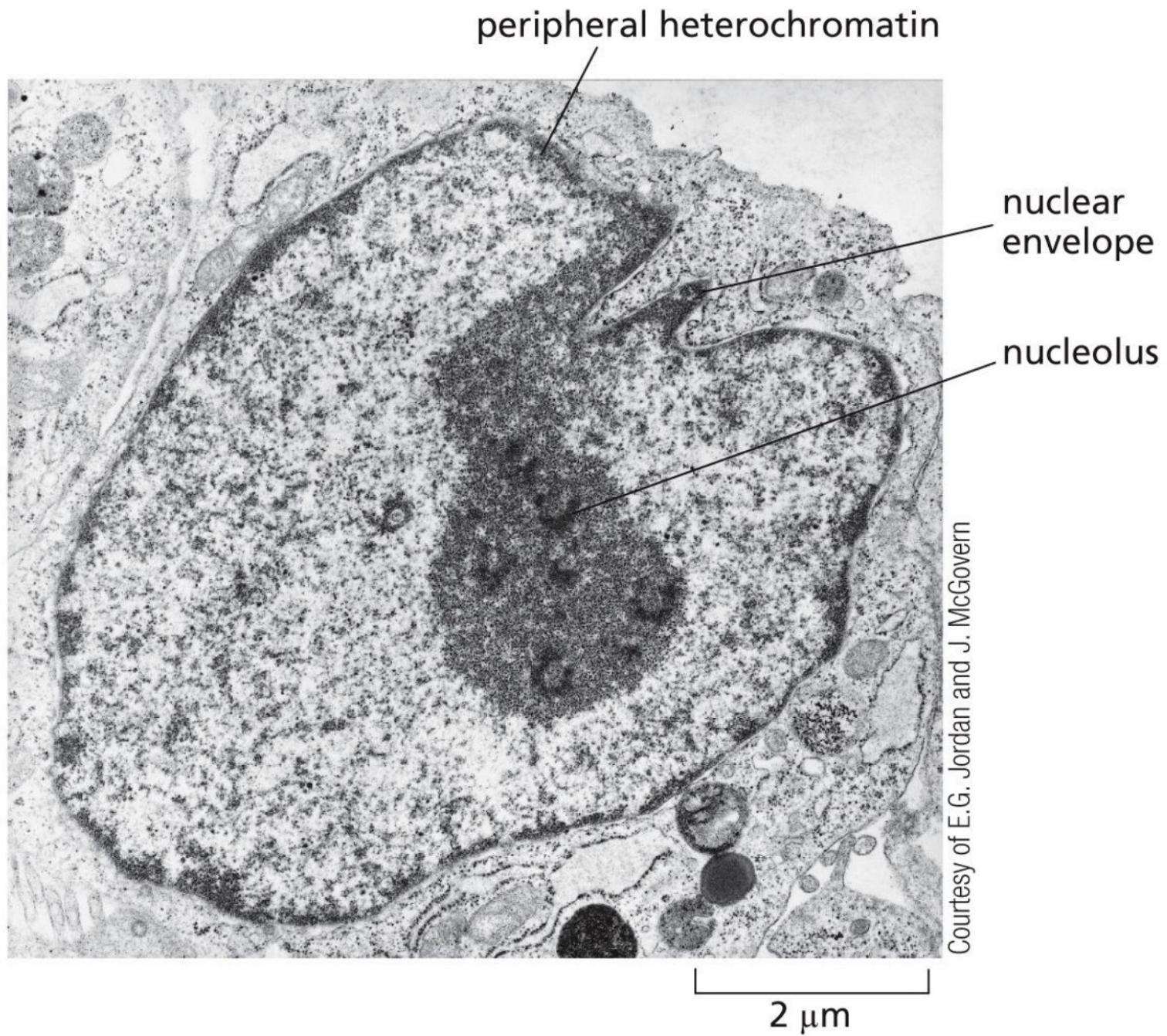
NUCLEUS



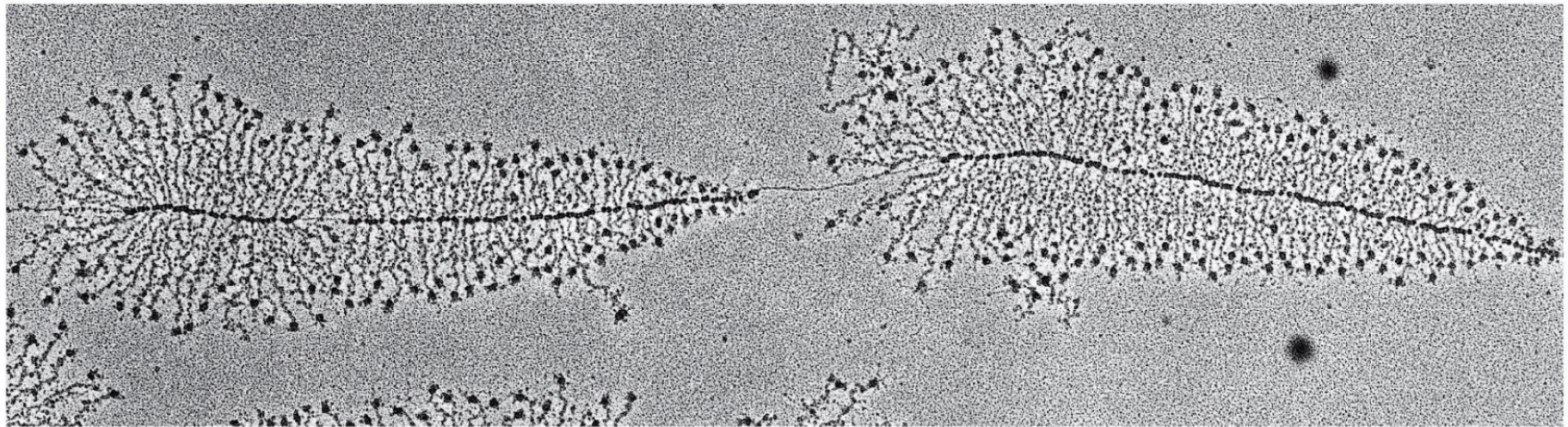
200 nm

Nuclear structures

What is a nucleolus?



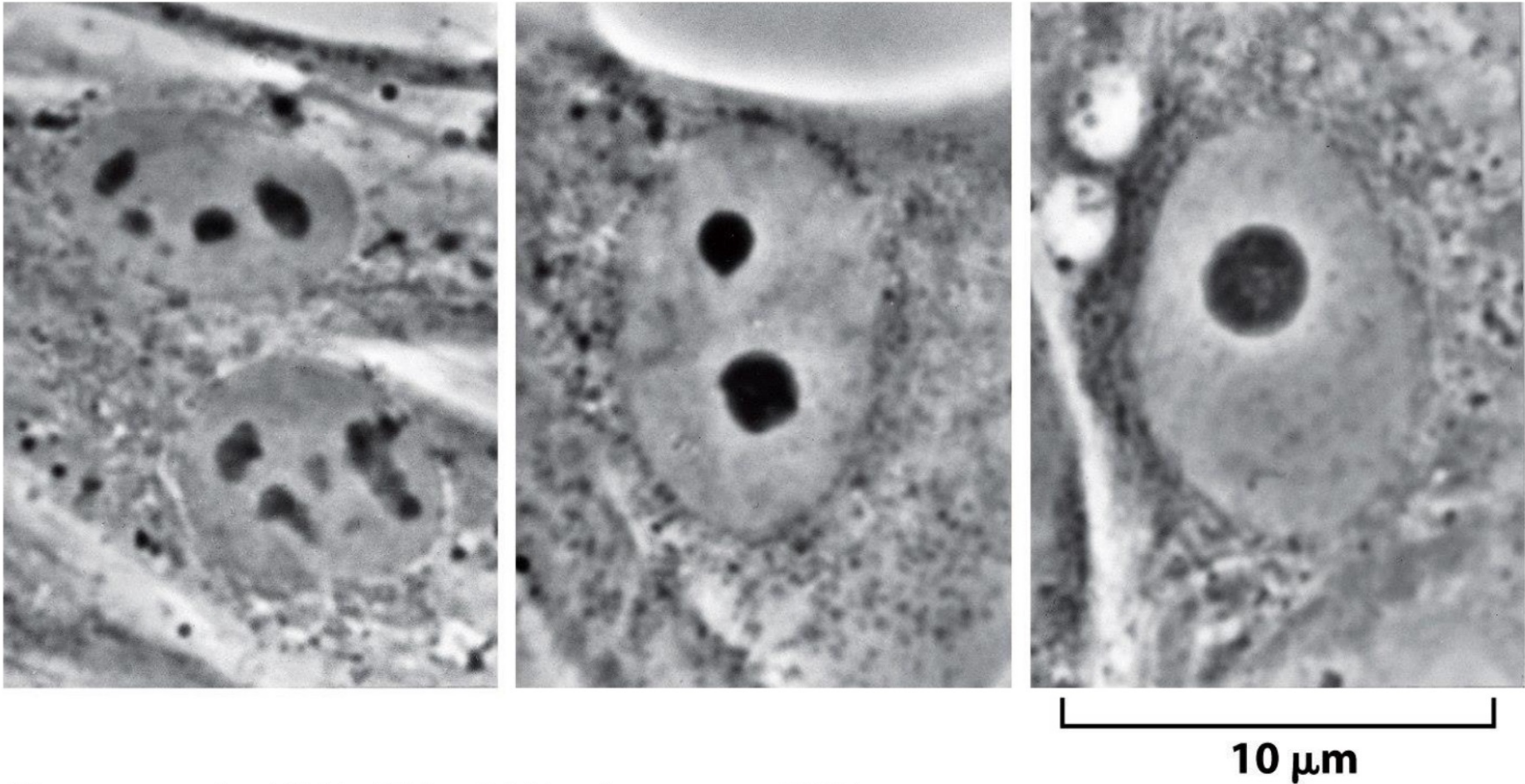
Example of transcription of rRNA genes



Courtesy of Ulrich Scheer

1 μm

**in the next
lecture:**

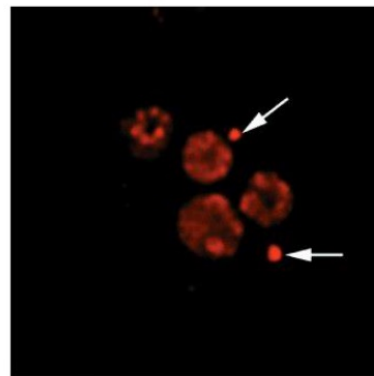


Here, not only rRNA, U6 snRNP, telomerase, tRNA

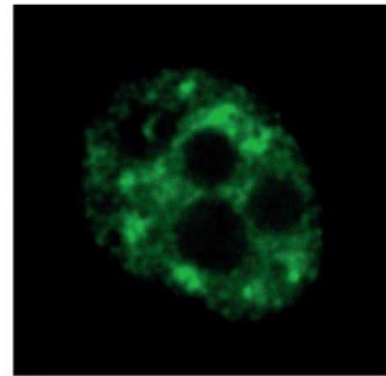
in the next lecture:

nucleoli and
Cajal bodies (arrows)

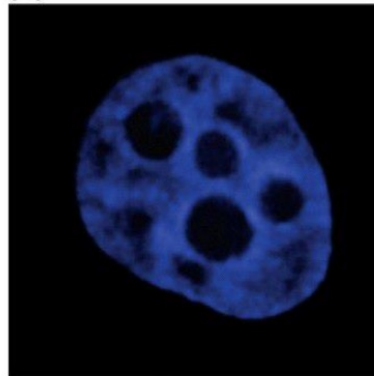
interchromatin granule
clusters/speckles



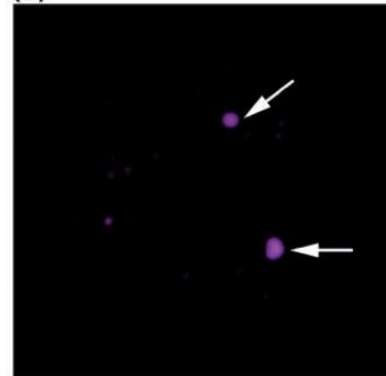
(A)



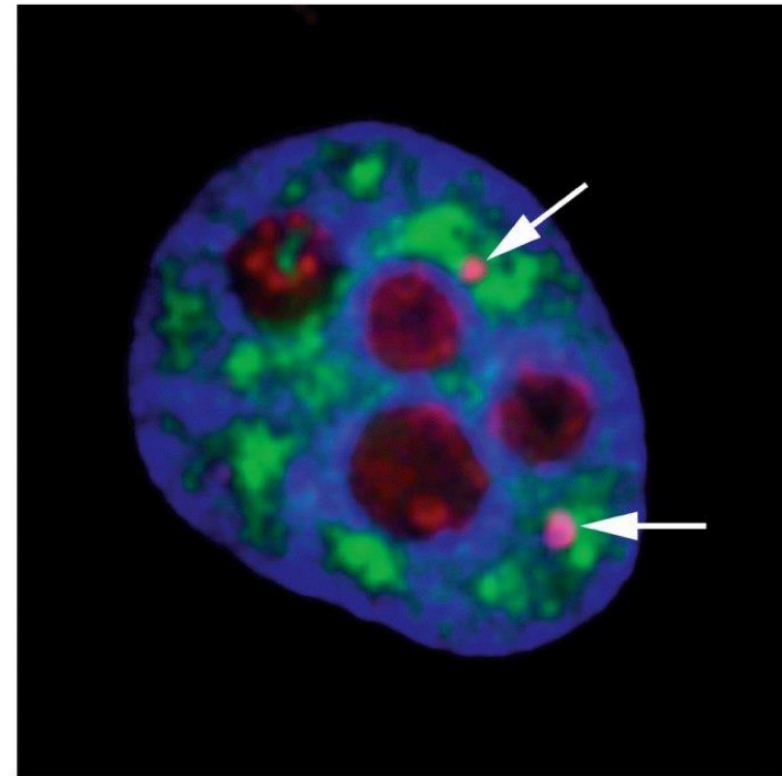
(B)



(C)



(D)



(E)

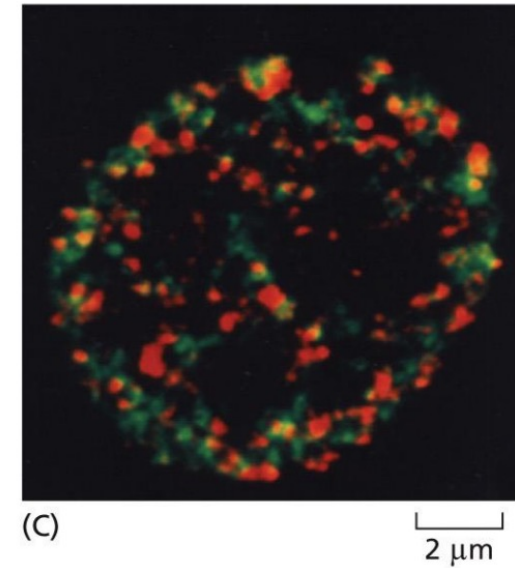
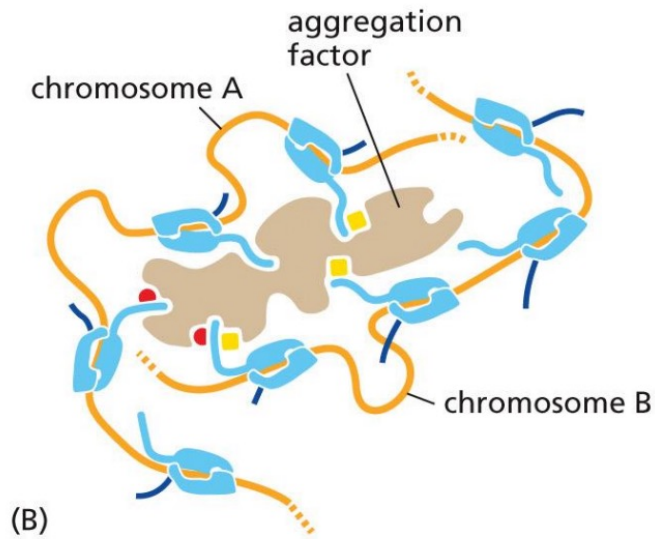
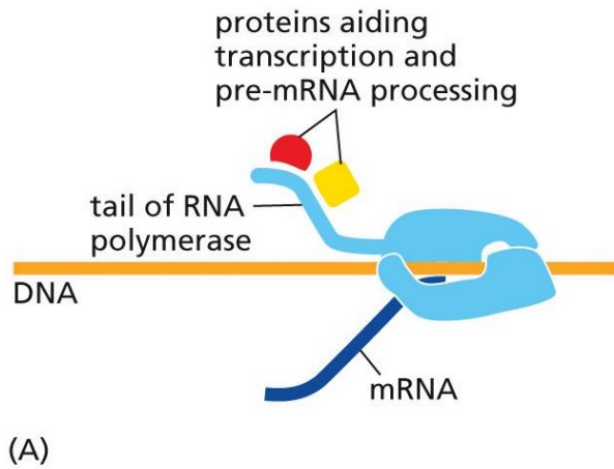
5 μm

bulk chromatin

Cajal bodies

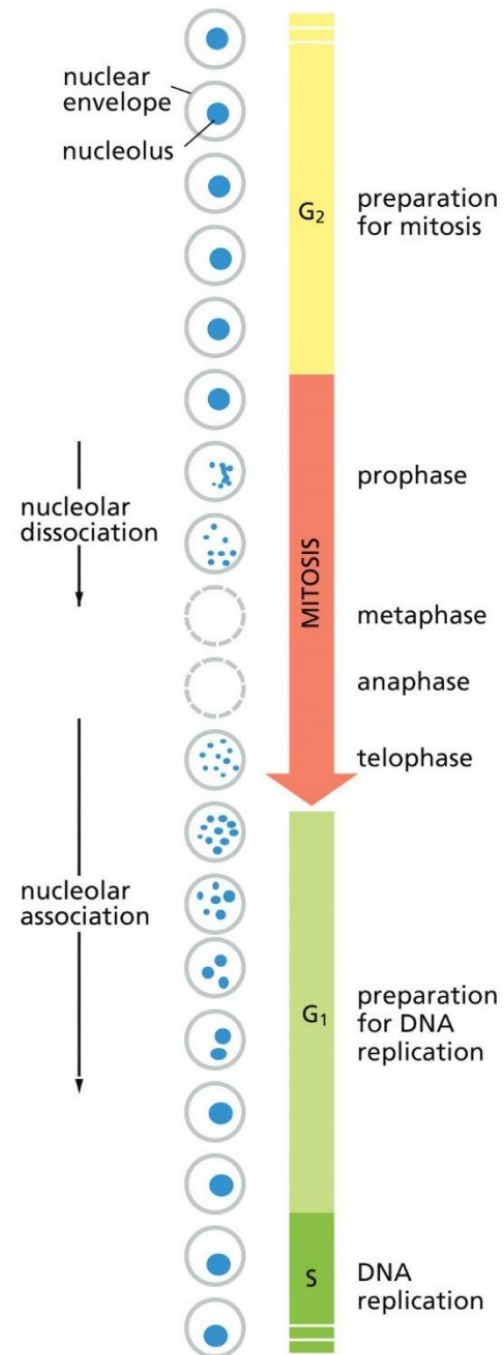
Cajal: snoRNA and snRNA
maturation/recycle/reset
Speckles: snRNP storage for splicing

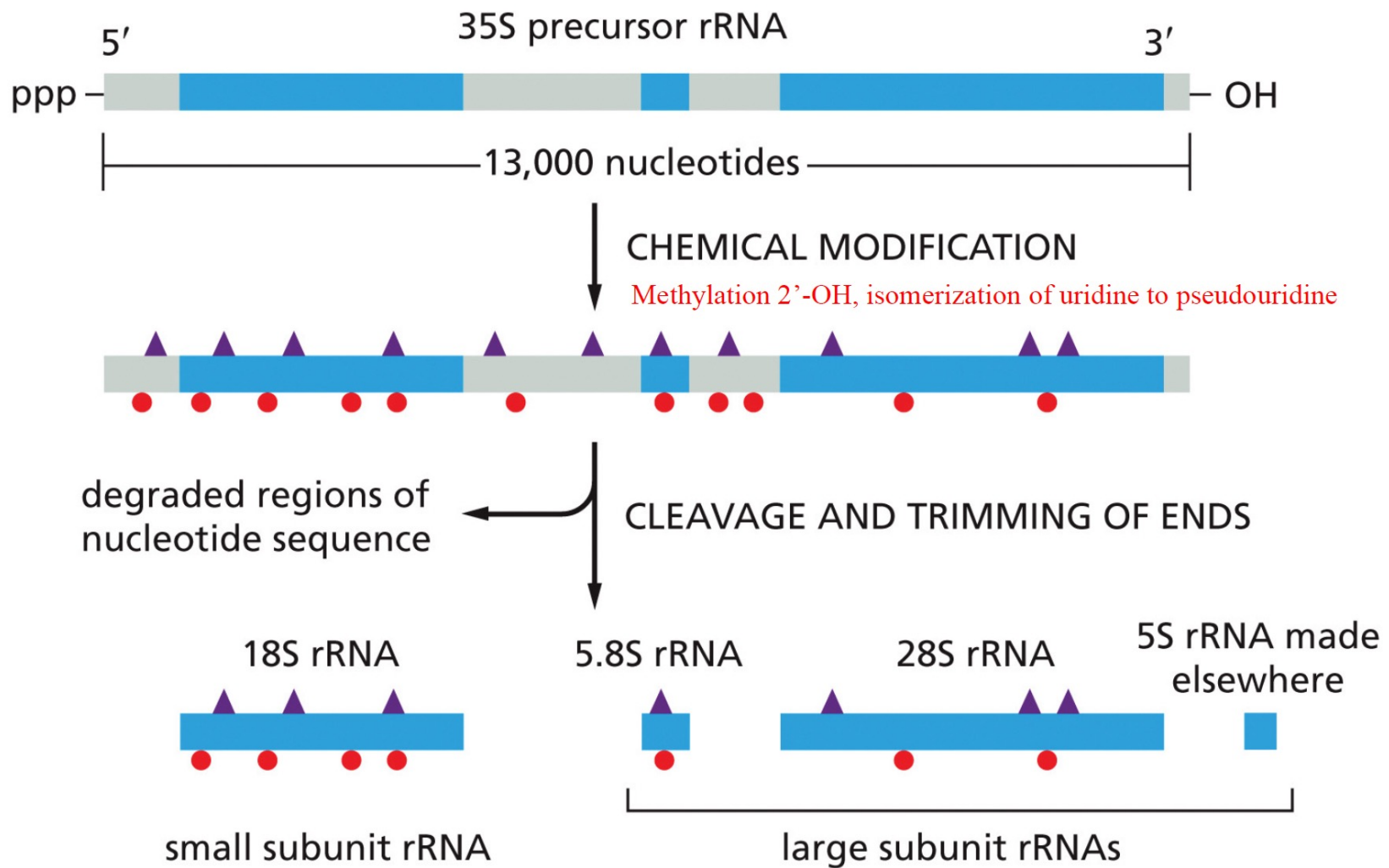
in the next lecture: Nuclear Structures: Transcription Factory



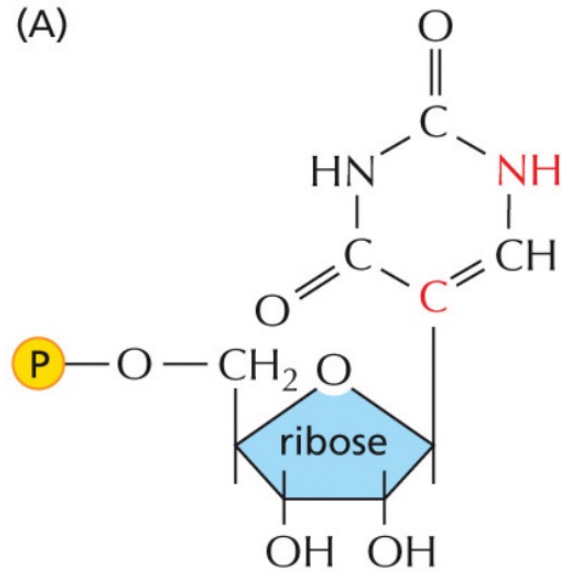
C, from D.G. Wansink et al., *J. Cell Sci.* 107:1449–1456, 1994. With permission from the Company of Biologists.

in the next lecture:

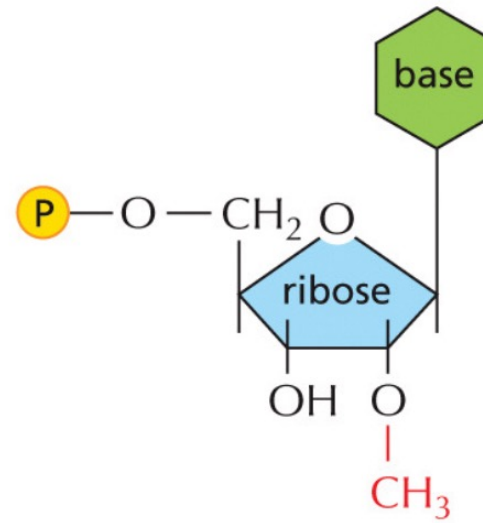




(A)

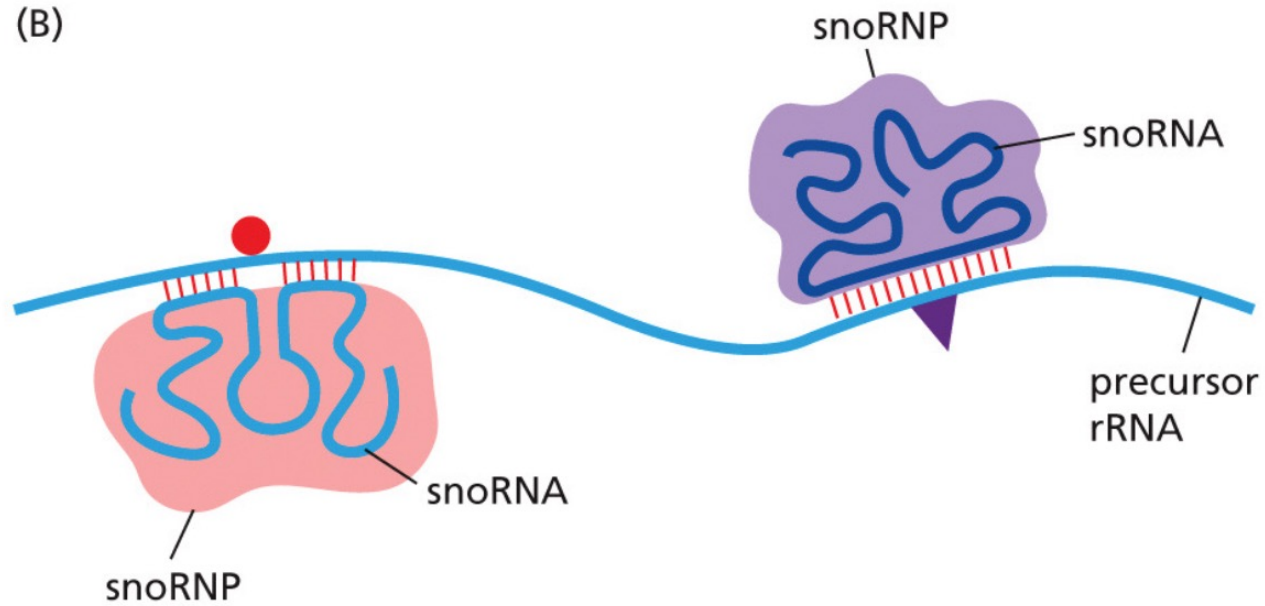


pseudouridine monophosphate



2'-O-methylated nucleotide

(B)



...Regulation of gene expression by non coding RNAs

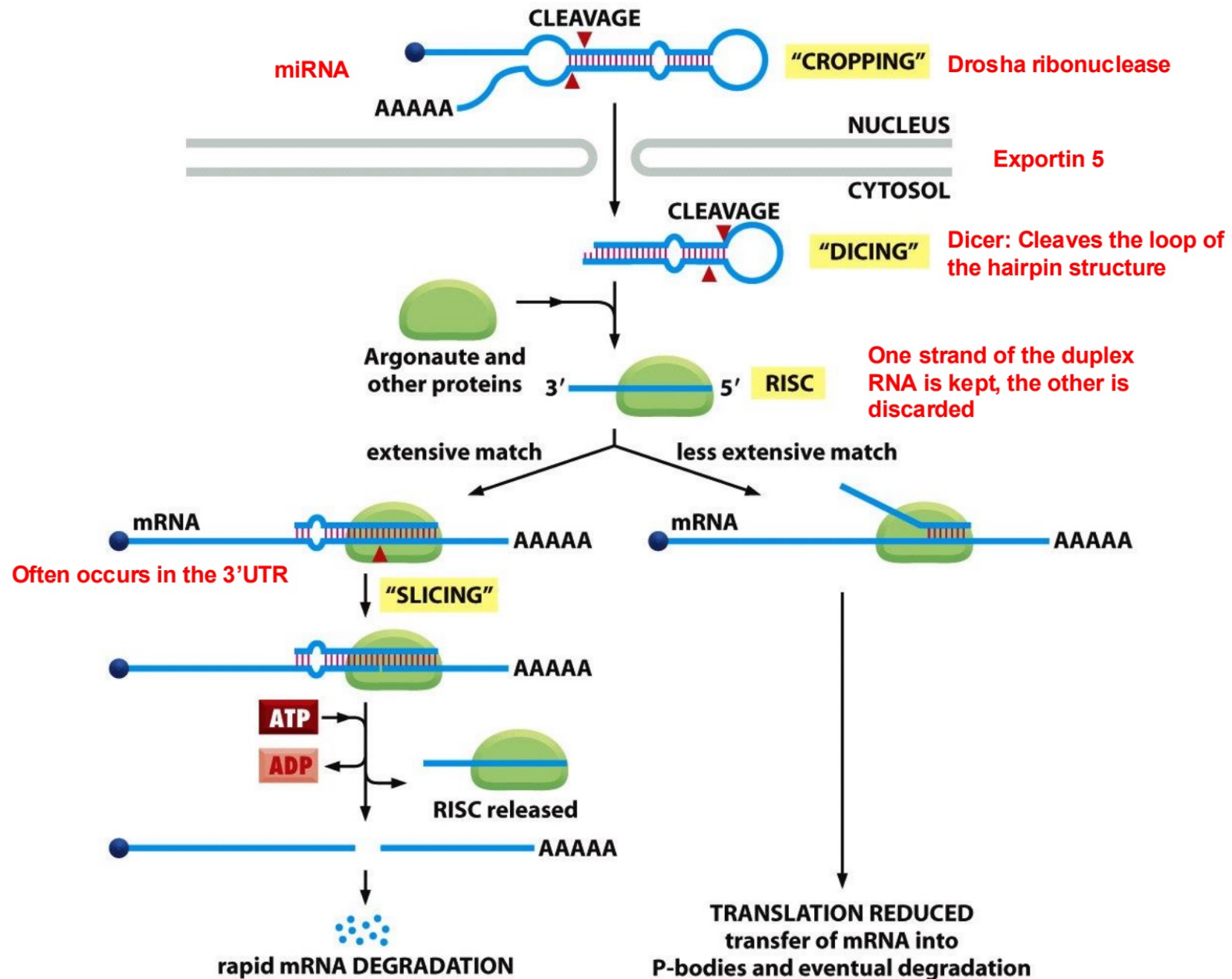


Figure 7-78 = Chapter 7

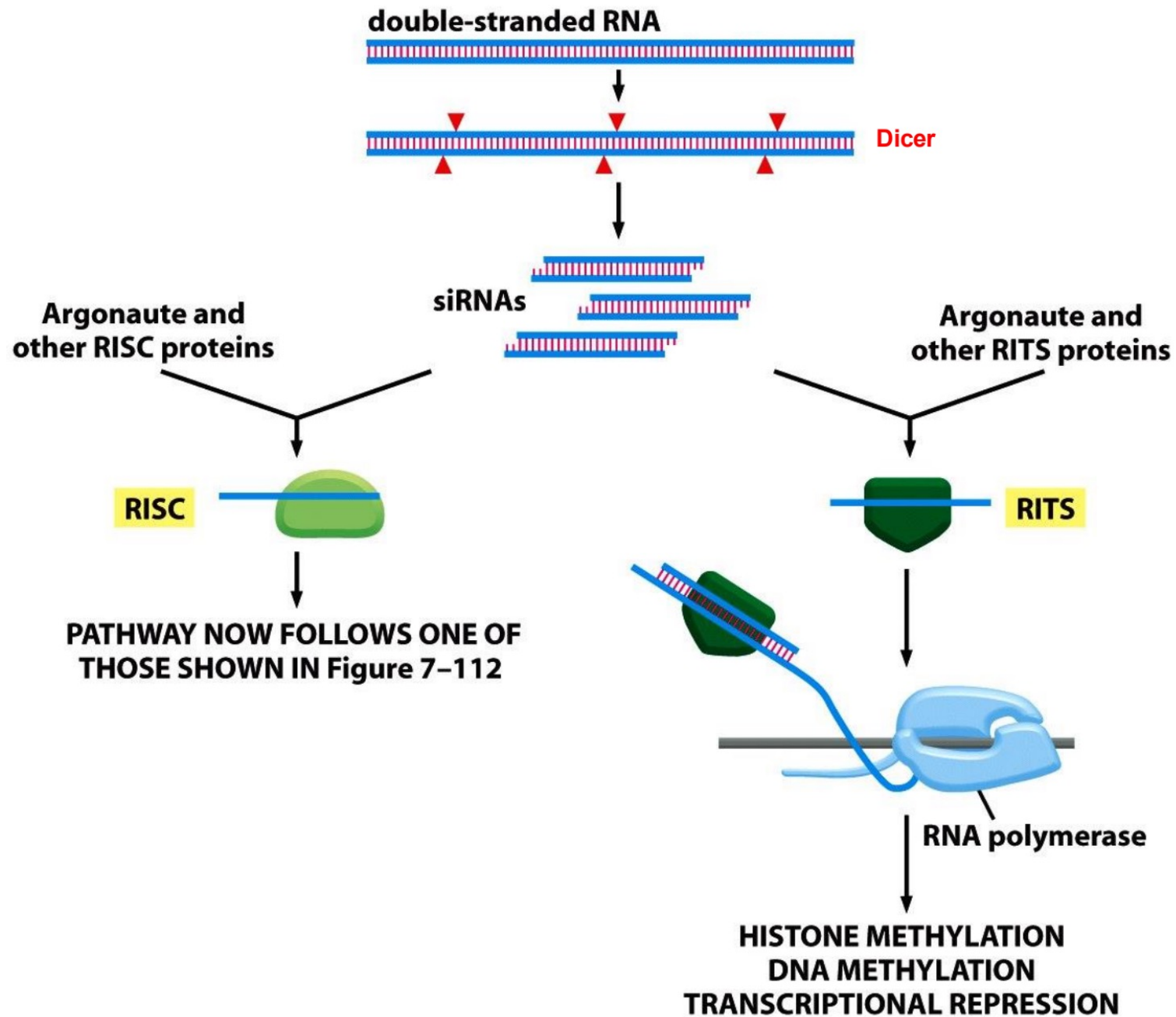


Figure 7-80 = Chapter 7

Take-home messages (most important concepts)

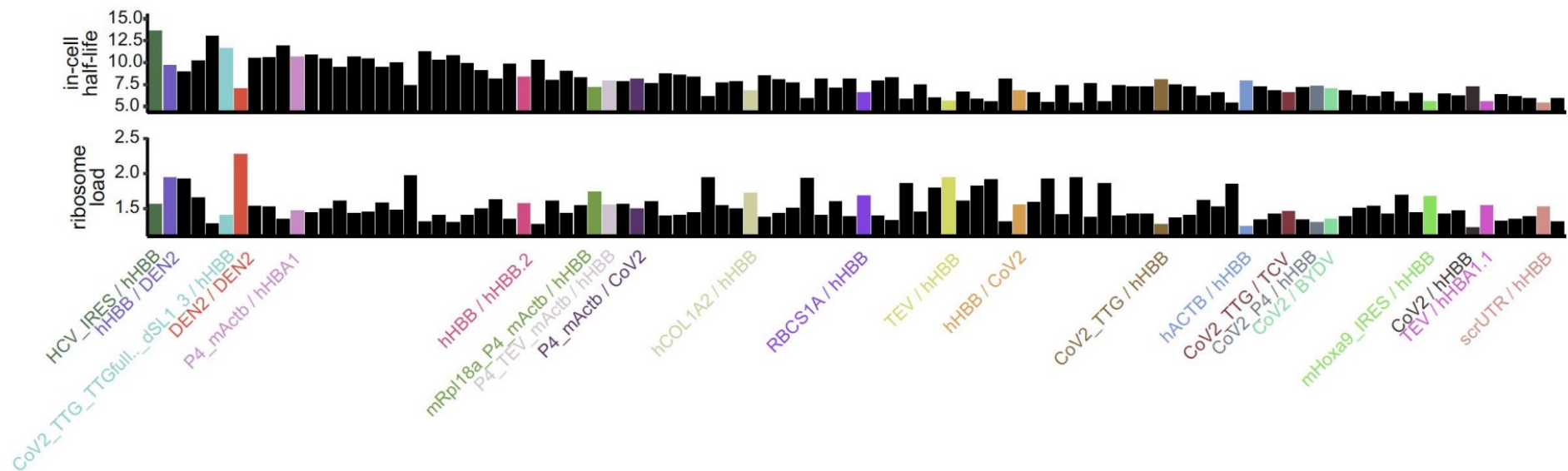
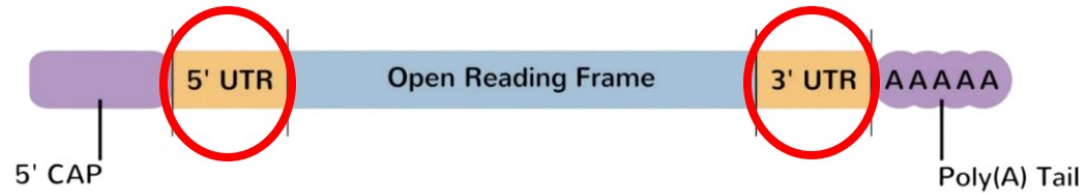
- RNA is the information-carrying molecule in gene expression;
- Ribosomes are the site of protein synthesis;
- mRNA can be seen as a short-lived DNA to protein intermediate;
- Transcription is always 5'→3' and catalyzed by RNA polymerases (highly processive, higher error rate than DNA Pol, and structurally distinct)
- Prokaryotes : one RNA Pol + σ factor; Eukaryotes: Pol I/II/III and general TFs (TFIID...H);
- TFIIF and the Pol II CTD coordinate RNA processing; Activators, Mediator, and chromatin-modifying complexes important for start of transcription, enhancer brought at the promoter by DNA looping;
- During elongation, elongation factors (as well as ATP-dependent chromatin remodelling factors) stabilize RNA Pol;
- Splicing in eukaryotes increases protein diversity (alternative splicing, enabled by “weak” consensus sites);
- During splicing, introns are removed as a lariat by the spliceosome (snRNAs U1/U2/U4/U5/U6); energy consuming process;
- Termination in Eukaryotic cells, CPSF, CstF, CF I/II, PAP etc. catalyze the 3' processing of mRNA;
- Nuclear organization: nucleolus (rRNA transcription/processing and ribosome assembly), Cajal bodies (snRNA/snoRNA maturation), and speckles (snRNP storage);
- Non-coding RNA regulate gene expression.

Practical example 1: mRNA structure, codon choice, UTRs choice influence mRNA stability and total protein output

mRNA vaccines: Modeling of the first year of vaccination estimates 14.4 million deaths averted based on reported COVID deaths, and ~19.8 million using excess-mortality estimates

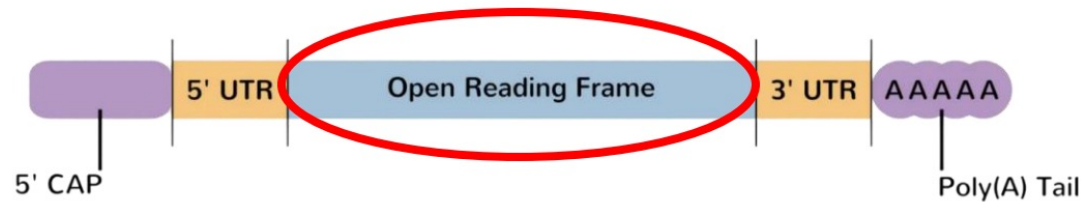


Practical example 1: mRNA structure, codon choice, UTRs choice influence mRNA stability and total protein output



Choice of UTRs affects in-cell half-life and ribosomal load (both affect total protein output)

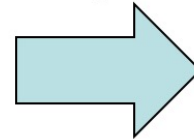
Practical example 1: mRNA structure, codon choice, UTRs choice influence mRNA stability and total protein output



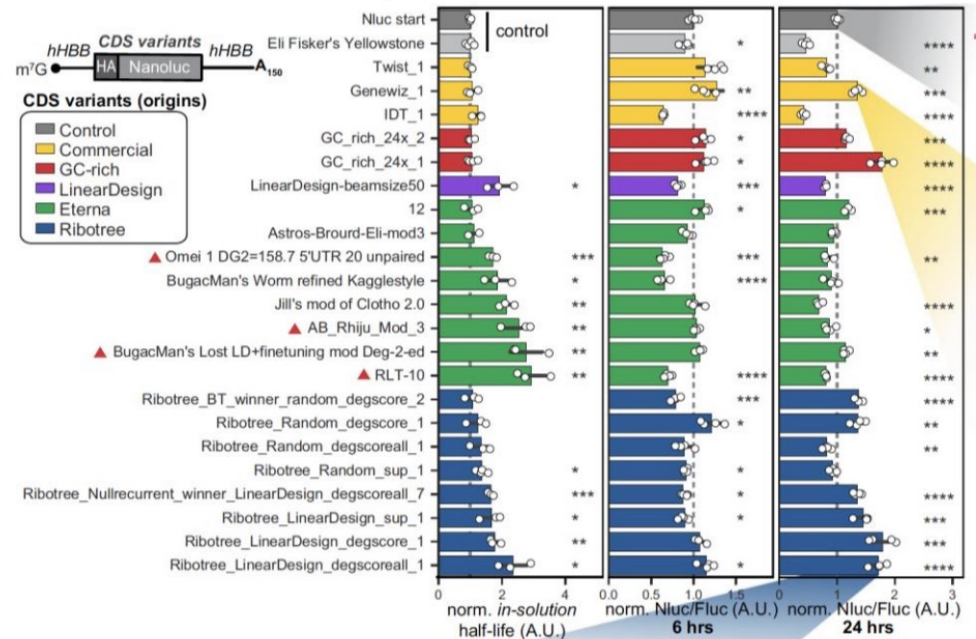
Human: (Genetic code: Standard)

Triplet	Amino acid	Fraction	Frequency/Thousand	Number	Triplet	Amino acid	Fraction	Frequency/Thousand
TTT	F	0.45	16.9	336562	TCT	S	0.18	14.6
TTC	F	0.55	20.4	406571	TCC	S	0.22	17.4
TTA	L	0.07	7.2	143715	TCA	S	0.15	11.7
TTG	L	0.13	12.6	249879	TCG	S	0.06	4.5
TAT	Y	0.43	12.0	239268	TGT	C	0.45	9.9
TAC	Y	0.57	15.6	310695	TGC	C	0.55	12.2
TAA	*	0.28	0.7	14322	TGA	*	0.52	1.3
TAG	*	0.20	0.5	10915	TGG	W	1.00	12.8
CTT	L	0.13	12.8	253795	CCT	P	0.28	17.3
CTC	L	0.20	19.4	386182	CCC	P	0.33	20.0
CTA	L	0.07	6.9	138154	CCA	P	0.27	16.7
CTG	L	0.41	40.3	800774	CCG	P	0.11	7.0
CAT	H	0.41	10.4	207826	CGT	R	0.08	4.7
CAC	H	0.59	14.9	297048	CGC	R	0.19	10.9
CAA	Q	0.25	11.8	234785	CGA	R	0.11	6.3
CAG	Q	0.75	34.6	688316	CGG	R	0.21	11.9
ATT	I	0.36	15.7	313225	ACT	T	0.24	12.8
ATC	I	0.48	21.4	426570	ACC	T	0.36	19.2
ATA	I	0.16	7.1	140652	ACA	T	0.28	14.8
ATG	M	1.00	22.3	443795	ACG	T	0.12	6.2
AAT	N	0.46	16.7	331714	AGT	S	0.15	11.9
AAC	N	0.54	19.5	387148	AGC	S	0.24	19.4
AAA	K	0.42	24.0	476554	AGA	R	0.20	11.5
AAG	K	0.58	32.9	654280	AGG	R	0.20	11.4
GTT	V	0.18	10.9	216818	GCT	A	0.26	18.6
GTC	V	0.24	14.6	290874	GCC	A	0.40	28.5
GTA	V	0.11	7.0	139156	GCA	A	0.23	16.0
GTG	V	0.47	28.9	575438	GCG	A	0.11	7.6
GAT	D	0.46	22.3	443369	GGT	G	0.16	10.8
GAC	D	0.54	26.0	517579	GGC	G	0.34	22.8
GAA	E	0.42	29.0	577846	GGA	G	0.25	16.3
GAG	E	0.58	40.8	810842	GGG	G	0.25	16.4

Codon choice affects total protein output (same protein is produce)

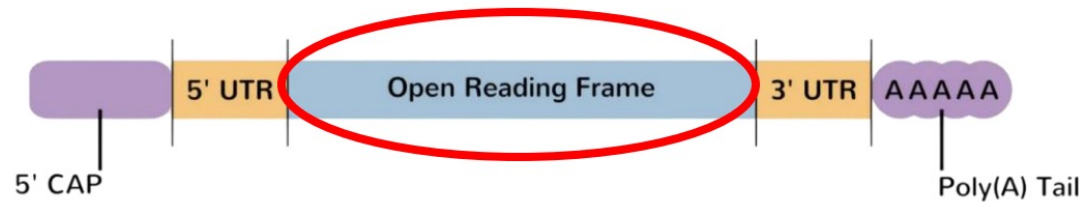


Codon optimization tools



Not all codons are present in equal amounts

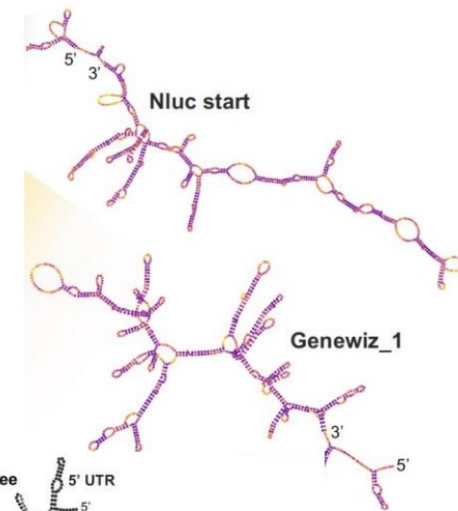
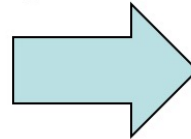
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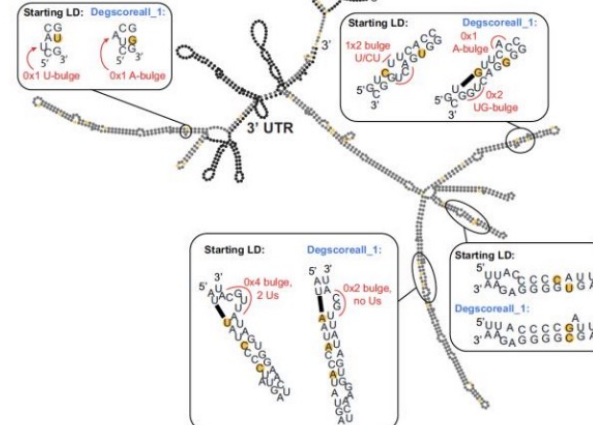
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Codon choice affects structure (same protein is produce)



DegScore improvement of RiboTree



Not all codons are present in equal amounts

Practical example 1: mRNA structure, codon choice, UTRs choice influence mRNA stability and total protein output

